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Illustrating long-run fiscal challenges using the Long-Term Model

Yvan Guillemette

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ABSTRACT/RÉSUMÉ

Illustrating long-run fiscal challenges using the Long-Term Model

This paper presents updated estimates of long-run fiscal gaps for 35 OECD and accession candidate countries using the OECD Global Long-Term Model. The long-run fiscal gap measures the fiscal adjustment required at the 2060 horizon to keep public debt ratios stable near current levels while maintaining current service standards and benefit entitlements. The methodology incorporates several refinements, including projected increases in defence spending and a more detailed treatment of primary expenditure. In the absence of additional policy action, the median country would require a fiscal adjustment of 4.7 percentage points of GDP between 2027 and 2060 to stabilise debt, with substantially larger adjustments in some countries. Population ageing is the main driver of long-run fiscal pressures through higher pension, health and long-term care spending, compounded in many cases by weak initial fiscal positions. The paper also assesses the sensitivity of fiscal gaps to key assumptions and policy scenarios.

Keywords: long-term projection, long-run fiscal gap, fiscal sustainability, fiscal pressure

JEL Classification: H68, E17, C53, O57, H55

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Illustration des défis budgétaires à longue échéance à l'aide du modèle à long terme

Ce document présente des estimations actualisées des écarts budgétaires à long terme pour 35 pays de l'OCDE et pays candidats à l'adhésion, fondées sur le modèle mondial de long terme de l'OCDE. L'écart budgétaire à long terme mesure l'ajustement budgétaire nécessaire à l'horizon 2060 pour stabiliser les ratios de dette publique à des niveaux proches de ceux observés actuellement, tout en maintenant les normes de service et les droits aux prestations en vigueur. La méthodologie intègre plusieurs améliorations, notamment la prise en compte des hausses projetées des dépenses de défense et un traitement plus détaillé des dépenses primaires. En l'absence de nouvelles mesures, le pays médian devrait procéder à un ajustement budgétaire de 4,7 points de pourcentage du PIB entre 2027 et 2060 pour stabiliser sa dette, des ajustements nettement plus importants étant requis dans certains pays. Le vieillissement démographique constitue le principal facteur de pression budgétaire à long terme, en raison de la hausse des dépenses liées aux retraites, à la santé et aux soins de longue durée, une situation aggravée dans de nombreux cas par une position budgétaire initiale fragile. Le document évalue également la sensibilité de ces écarts budgétaires aux principales hypothèses et aux différents scénarios de politique publique.

Mots-clés : projection à long terme, écart budgétaire à long terme, viabilité des finances publiques, pression budgétaire

Classification JEL : H68, E17, C53, O57, H55

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Illustrating long-run fiscal challenges using the Long-Term Model

By Yvan Guillemette[1]

Main findings

This paper presents the latest estimates of long-run fiscal gaps based on the OECD Global Long-Term Model. It describes the current methodology in detail and assesses the indicator's sensitivity to key assumptions, covering 35 OECD and accession candidate countries. This version introduces several refinements to the previous methodology, including explicit accounting for projected increases in defence expenditure, a more detailed treatment of current primary expenditure, and the inclusion of revaluation effects on government financial assets. The analysis provides illustrative estimates of future fiscal challenges across countries using a common stylised framework. Given this, individual-country results should not be viewed as precise quantifications. The main findings are:

- In the absence of further policy changes, maintaining current service standards and benefits while keeping the gross public debt ratio stable would require a fiscal adjustment of 4.7 percentage points (pp) of GDP between 2027 and 2060 in the median country. In ten countries, the required fiscal adjustment exceeds 8 pp of GDP.
- The main sources of long-run fiscal gaps are weak initial fiscal positions relative to those that would keep government debt-to-GDP ratios stable (0.9 pp of GDP in the median country), and population ageing, which raises public expenditure on both pensions (2.2 pp) as well as health and long-term care (1.8 pp).
- Substantial long-run fiscal gaps imply the need for higher revenue or commensurate expenditure restraint to prevent rising government debt ratios. Without revenue adjustments or spending reductions, government net debt could rise by 43 pp of GDP by 2045 in the median country, and by up to 124 pp in the United States.
- The required fiscal adjustment through 2060 is marginal (<0.5 pp of GDP) or negative in three countries — Denmark, Iceland and Ireland — reflecting relatively strong initial fiscal positions

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as well as low or negative net public debt, and in the case of Denmark future statutory retirement ages linked to life expectancy.

- Long-run fiscal gap estimates are sensitive to interest-rate assumptions. In a scenario with a one percentage point increase in interest rates over baseline but unchanged growth, long-run fiscal gaps rise by 1-1¼ pp of GDP in countries with the largest debt burdens, such as Greece, Italy, and Japan.
- Boosting growth could help improve fiscal outcomes but is unlikely to fully offset long-run adjustment needs, and the fiscal gain depends on the source of growth. Reforms that raise average annual potential growth by 0.3 pp through to 2060 have little impact on long-run fiscal gaps if they work by increasing economy-wide trend labour efficiency. This is because much government expenditure is made up of wages and transfers whose real cost tends to rise with broad productivity growth. Reforms that operate through higher employment have a substantial effect on long-run fiscal gaps (2½ pp reduction relative to baseline in the median country). The fiscal dividends from higher growth could be larger if interest rates do not adjust one-for-one to higher growth rates in the long run, if efficiency gains are larger in the public than in the private sector, or if tax thresholds are indexed to prices instead of wages.
- Increasing net inward international migration by 0.1% of the population by 2033 could reduce long-run fiscal gaps by about 0.6 pp of GDP in the median country by 2060 due to the younger age composition of migrants, with the total population about 3½ per cent higher than baseline. Most of this fiscal dividend stems from lower pension spending, as higher migration reduces the ratio of retirees to workers.
- Raising the statutory retirement age by one year lowers long-run fiscal gaps by 0.4 pp of GDP in the median country, with even larger effects where pension expenditures are relatively high and/or where retirees-to-workers and population-to-workers ratios are projected to rise the most.
- Shifting from the baseline assumption of full wage indexation to price indexation of pension benefits has a substantial effect, reducing long-run fiscal gaps by 4½ pp of GDP in the median country by 2060, and by 7-9 pp in the most affected countries, although this would imply an increase in old-age poverty.
- In an energy-transition scenario consistent with the Paris Agreement goal of limiting global temperature increases to well below 2°C by 2100, substantial capital investment in new electricity-generation capacity would be required. If governments finance a large share of this investment, the transition could add a few percentage points of GDP to expenditure pressure over the next decade in some countries.

1. Introduction

Most OECD governments face long-run fiscal challenges due to weak initial fiscal positions, rising financing costs, slowing economic growth, and mounting spending pressures linked to population ageing and the increasing relative price of government-provided services. The magnitude of this challenge can be illustrated with long-run fiscal gap estimates from the OECD Global Long-Term Model (LTM).

This indicator, previously called “long-run fiscal pressure”, first appeared in Guillemette and Turner (2018[1]). The methods for projecting certain fiscal components and the way the overall long-run fiscal gap is decomposed have evolved subsequently, for instance separating out defence expenditure and projecting it explicitly. In addition to presenting the latest estimates, this paper describes the current methodology in detail and assesses the indicator’s sensitivity to key assumptions. It also updates estimates of capital costs for new electricity generation previously published in Guillemette and Château (2023[2]), a

potential additional fiscal challenge not included in the standard indicator. Although the LTM horizon is now 2100, the analysis in this note runs to 2060 to be comparable to earlier work.

The scenarios and fiscal projections in this note are intended to illustrate broad trends and support discussion of future fiscal challenges. They aim to provide approximate orders of magnitude and, by applying a common and relatively simple framework across countries, enable meaningful cross-country comparisons. However, the results should not be viewed as precise estimates for individual countries. The LTM relies largely on stylised assumptions and reduced-form equations and omits many country-specific factors, a necessary trade-off for generating scenarios across numerous economies. Such factors may include country-specific institutional arrangements or projected balance-sheet developments, such as the accumulation of assets in funded public pension plans. The LTM also does not account for potential shorter-term macroeconomic responses to fiscal actions. Moreover, uncertainty surrounding multi-decade projections is inherently large. Some sources of uncertainty are illustrated below, but the true extent, including unquantifiable risks, is likely to be substantially greater.

Broader analyses of long-run fiscal challenges and policy implications can be found in Rouzet et al. (2019[3]), Crowe et al. (2022[4]), Koutsogeorgopoulou and Morgavi (2025[5]) and OECD (2025[6]). In addition, OECD Economic Surveys provide extensive country-specific analysis of short and long-run fiscal challenges and policy recommendations. For instance, the potential short-run impact of sustained fiscal action is discussed in some recent Country Surveys (OECD, 2024[7]; OECD, 2024[8]).

The note is structured as follows. Section 2 presents the latest estimates of long-run fiscal gaps. Section 3 examines the sensitivity of these estimates to major assumptions underlying the long-term scenarios. Section 4 considers long-run fiscal sustainability from a different angle by projecting government debt-to-GDP ratios and gross financing needs. Finally, Section 5 updates previously published estimates of potential capital costs associated with building new electricity-generation capacity in both the baseline and an energy-transition scenario.

2. The long-run fiscal gap indicator and its components

The long-term model and the fiscal closure rule

The LTM is a country-level model with an annual frequency, comprising dynamic projection rules and identities for key variables over a horizon to 2100. For each country, the model's core is a production function with standard inputs — labour and capital — and labour efficiency. The LTM covers 139 individual countries, with a further 55 countries grouped into three regional aggregates. Most variables are anchored to the latest OECD Economic Outlook; for this note, the starting point is the OECD Economic Outlook 119 database of June 2026, with projections until 2027.

The model includes a "fiscal block" that projects government primary expenditure by function, interest payments and receipts, and the evolution of government financial assets and liabilities. This block is available for 33 OECD countries (Chile, Colombia, Costa Rica, Mexico and Türkiye are excluded due to incomplete comparable fiscal data) and three accession candidate countries (Bulgaria, Croatia, and Romania). Norway is typically excluded from cross-country comparisons because of its exceptionally large net financial asset position, although a modified fiscal block is available for more tailored analyses. This note therefore focuses on the remaining 35 countries for which a fiscal block exists in the LTM. Future work could aim to expand fiscal projections and the long-run pressure indicators to a larger number of countries.

In a multi-decade model, a technical requirement is that the government debt-to-GDP ratio must eventually stabilise. For this purpose, the model includes a reaction function that dynamically adjusts the ratio of cyclically adjusted primary revenue to potential GDP to stabilise the gross debt ratio at its value in the final

year of the OECD Economic Outlook projection (2027). This also implies stabilising the net government debt-to-GDP ratio given the LTM assumption that government financial assets remain stable as a share of GDP (see section 2.4). The projected change in the revenue ratio over a long horizon is referred to as the long-run fiscal gap or, interchangeably, the required long-run fiscal adjustment. It provides a benchmark metric of the fiscal challenge over the long run, accounting for the initial fiscal position as well as projected growth, interest rates, and expenditure trajectories.

The use of a revenue measure for debt stabilisation in the LTM is a technical modelling choice, not a policy prescription. The appropriate policy mix will differ across countries and may include revenue-raising measures alongside expenditure reforms, pension and health reforms, growth-enhancing reforms, and in some cases additional debt financing. Other considerations not captured in the LTM framework, such as distributional impacts, may also inform policy choices.

The long-run fiscal gap indicator has several advantages:

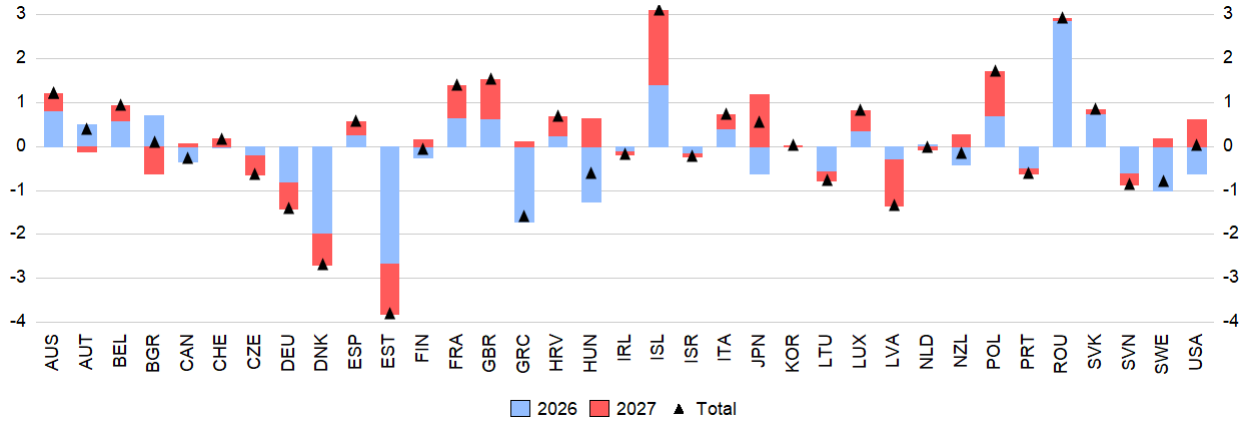
- It is defined in structural terms, adjusting for cyclical or temporary components in initial fiscal positions, despite uncertainties in such estimates.
- It is forward-looking, accounting for projected interest rate and growth trajectories.
- It avoids categorising fiscal positions as definitively sustainable or unsustainable — an assessment that cannot be made reliably given the inherent uncertainty of the future (Blanchard, Leandro and Zettelmeyer, 2021[9]).

The LTM approach allows very different government debt ratios across countries to persist. Alternative approaches could allow some increase in debt ratios for the lower-debt countries, or enforce reductions in debt ratios for highly indebted ones. In either case, the medium-term fiscal adjustment path would differ, but the size of the long-run adjustment would be little affected (see section 3.2). Another potential issue is that the long-run fiscal gap considers only the impact of population ageing on the spending side of the fiscal ledger, but not the revenue side. Recent OECD research has shown that population ageing could depress the tax-to-GDP ratio by 1 percentage point on average by 2060, mostly via a reduction in labour income taxes (Sicsic and Hourani, 2026[10]). Although modest relative to the impact of ageing on spending, such changes represent an additional long-term fiscal gap that may need closing with policy measures such as base broadening or tax rate changes.

The LTM-based fiscal adjustment begins in 2028 and comes in addition to any projected fiscal adjustment in 2026 and 2027 in the OECD Economic Outlook 119 database. The adjustment between 2025 and 2027 is small in the median country but exceeds ½ per cent of GDP in 13 countries (Figure 1). Thereafter, an annual cap of one percentage point of potential GDP is imposed on the fiscal adjustment to reflect political-economy constraints. While the one-percentage-point limit is somewhat arbitrary, larger sustained adjustments have been comparatively rare. The constraint implies that it can take a few years for the objective of a stable debt ratio at target to be achieved in countries where the initial fiscal position is very far from the level required to stabilise the debt ratio.

Figure 1. Projected change in cyclically-adjusted government primary balance, 2025-27

% pts of potential GDP



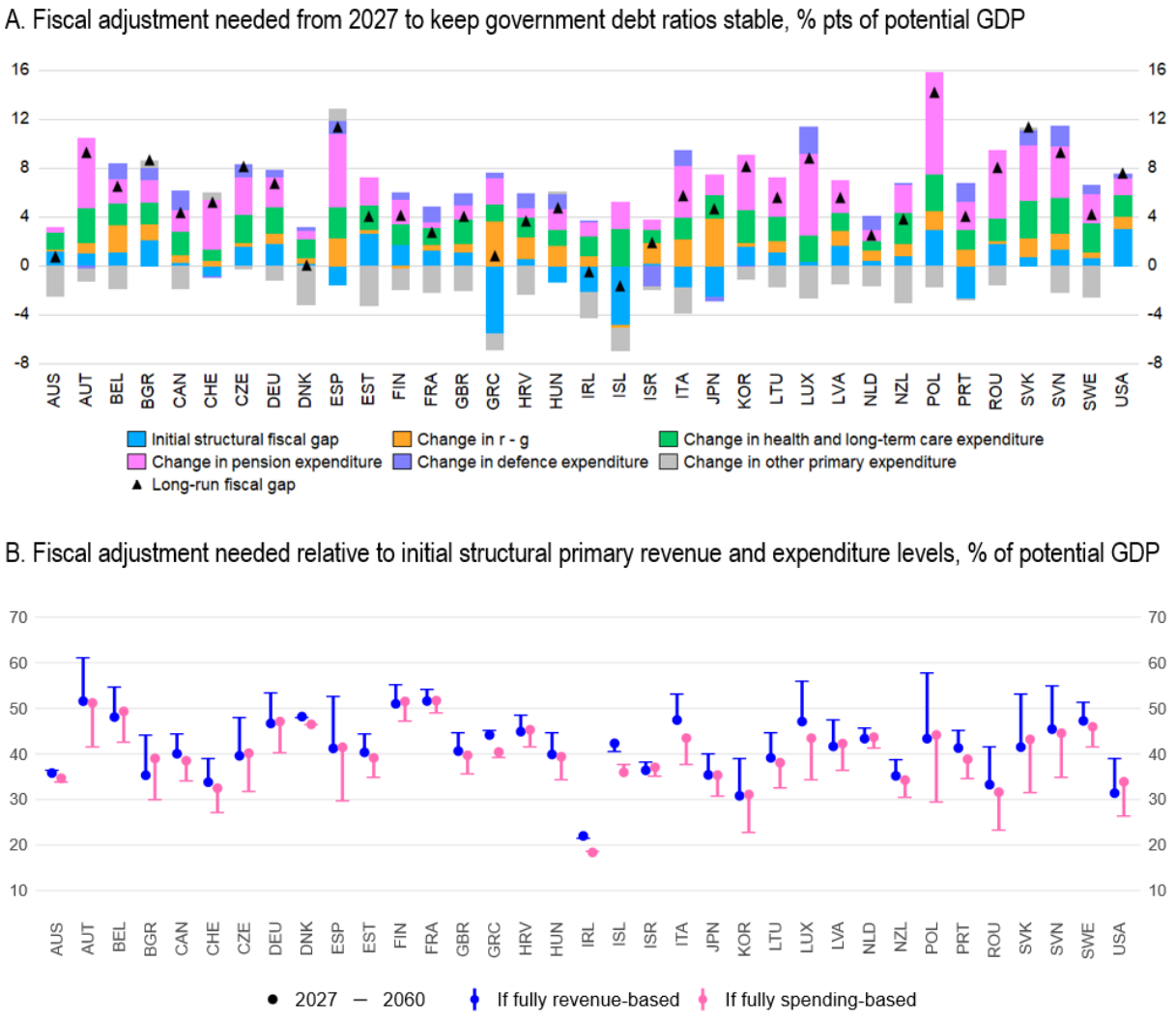
Note: The change in the cyclically-adjusted primary government balance serves as proxy for the degree of structural fiscal adjustment.
Source: OECD Economic Outlook 119 database.

The LTM-based fiscal trajectories and the annual cap on the size of the fiscal adjustment may diverge from current government plans and may differ from national or supranational fiscal objectives, targets, or rules. Moreover, the model does not account for potential short-term demand effects of fiscal consolidation or for negative supply-side feedback from higher taxation. The gap between actual and cyclically-adjusted revenue is determined by the output gap — assumed to close gradually over a few years — and by a country-specific semi-elasticity coefficient estimated in earlier OECD work (Price, Dang and Botev, 2015[11]).

Long-run fiscal gaps

In the standard business-as-usual scenario, the long-run fiscal gap is 4.7 percentage points (pp) of GDP in the median country at the 2060 horizon (Figure 2). In ten countries, the gap exceeds 8 pp. It is marginal (<0.5 pp) or negative in only three countries — Denmark, Iceland and Ireland — reflecting relatively strong initial fiscal positions as well as low or negative net public debt, and in the case of Denmark future statutory retirement ages that are linked to life expectancy. The time profile of the fiscal gap for the median country is somewhat front-loaded, with about 2.6 of the 4.7 pp long-run fiscal adjustment being implemented in the first decade of the projection period, in large part because of a median initial structural fiscal gap of nearly 1 pp of GDP (Figure 3).

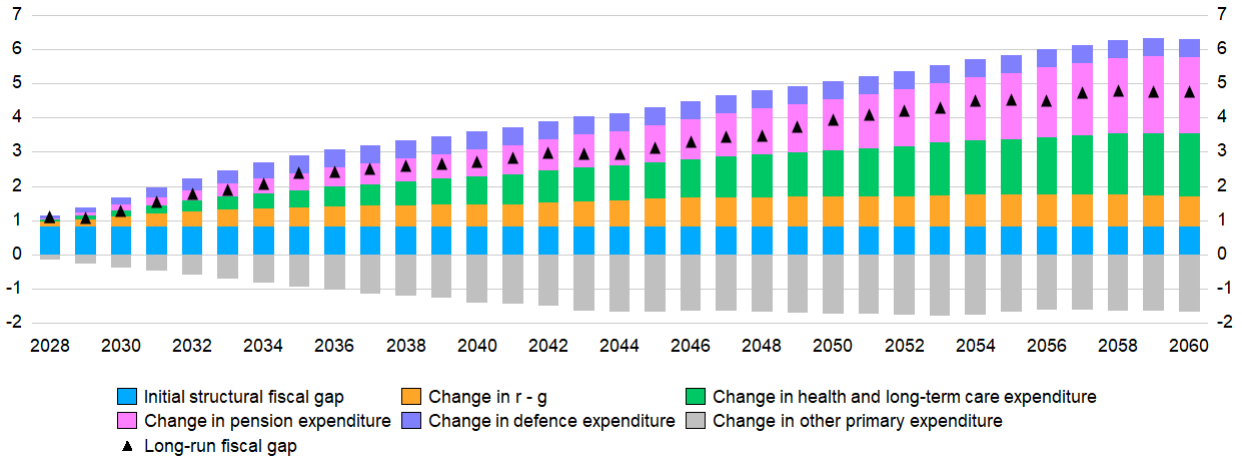
Figure 2. Long-run fiscal gap at the 2060 horizon



Note: The charts show the fiscal adjustment required over the 2027-to-2060 period to keep the gross debt-to-GDP ratio stable near its current value (which also implies a stable net debt-to-GDP ratio given the assumption that government financial assets remain stable as a share of GDP). The underlying projected growth rates, interest rates, etc., are from the BAU1 scenario described in OECD (2025[12]) but updated to use the OECD Economic Outlook 119 database as the starting point. In Panel A, the required fiscal adjustment is decomposed into component parts explained in the main text and decomposed algebraically in Annex A. In Panel B, it is shown as a segment length relative to the 2027 structural primary revenue level (if the adjustment is fully revenue-based) or expenditure level (if fully spending-based).
Source: OECD Economic Outlook 119 long-term scenarios database.

Figure 3. Evolution of fiscal gap in the median country

Fiscal adjustment needed from 2027 to keep government debt ratios stable, % pts of potential GDP



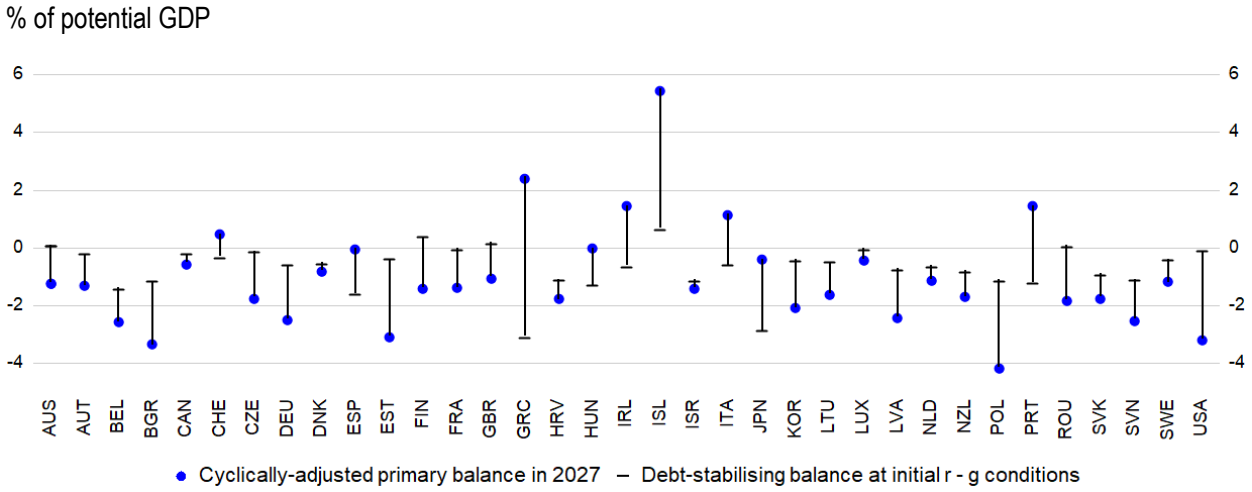
Note: The charts show the fiscal adjustment required from 2027 to the year given by the x-axis to keep the gross debt-to-GDP ratio stable near its current value (which also implies a stable net debt-to-GDP ratio given the assumption that government financial assets remain stable as a share of GDP). The underlying projected growth rates, interest rates, etc., are from the BAU1 scenario described in OECD (2025[12]) but updated to use the OECD Economic Outlook 119 database as the starting point. The required fiscal adjustment is decomposed into component parts explained in the main text and decomposed algebraically in Annex A. Because medians are not additive, the diamond does not necessarily correspond to the sum of components.
Source: OECD Economic Outlook 119 long-term scenarios database.

The long-run fiscal gap can be decomposed into several components with intuitive interpretations (see Annex A for the algebraic decomposition). The following subsections look at each in turn, decomposing its drivers where feasible.

The initial structural fiscal gap

The initial structural fiscal gap measures the difference between the structural primary balance in 2027 and that which would stabilise the debt-to-GDP ratio at its 2027 level at the growth and interest rates projected to prevail that year. It is 0.9 pp in the median country, though variation across countries is substantial. Four countries, including the United States, show an initial structural fiscal gap of 2 pp of GDP or more (Figures 2 and 4).

Figure 4. Initial structural fiscal gap



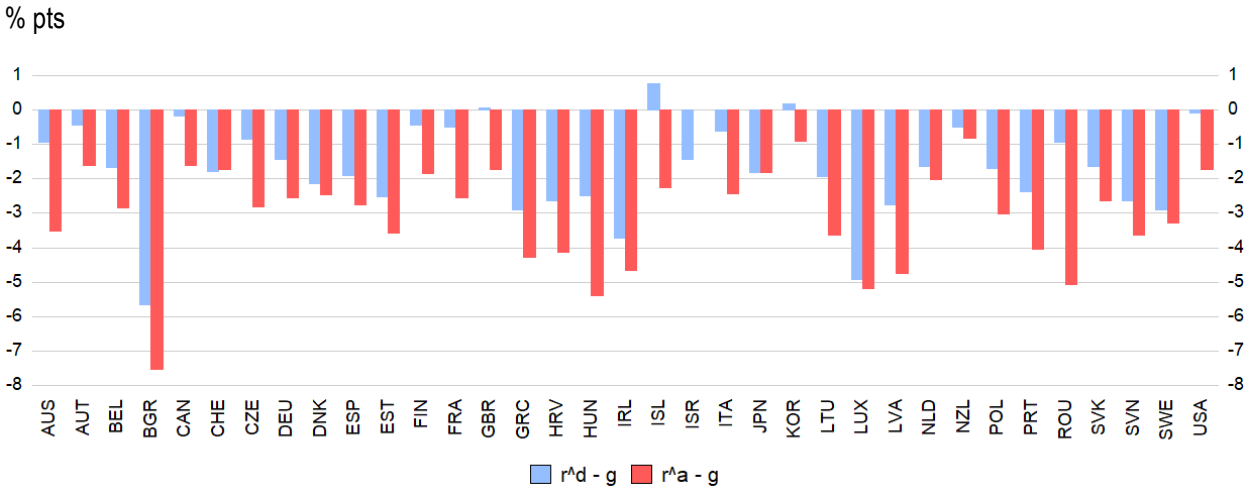
Note: The size of the initial structural fiscal gap is represented by the length of the vertical segment.
Source: OECD Economic Outlook 119 long-term scenarios database.

The size of the initial structural fiscal gap depends on initial-year values for several variables (see Appendix A for the algebraic definitions):

- The cyclically-adjusted primary budget balance. A higher balance implies a smaller gap, all else equal.
- The levels of gross government debt and financial assets.
- The differential between the implicit average interest rate paid on government debt (r^d) and the economy's potential growth rate (g) — the so-called $r - g$ differential. This differential is central in any long-run fiscal projection because it determines whether the debt-to-GDP ratio would rise, stabilise, or fall for a given primary balance. Conditional on a positive stock of government debt, a higher $r^d - g$ differential increases the initial fiscal gap.
- The differential between the implicit average return on government financial assets (r^a) and potential growth (g). With positive asset holdings, a higher $r^a - g$ differential reduces the initial fiscal gap.

The contributions of initial $r - g$ differentials and those of debt and asset stocks are interdependent, making a clean decomposition infeasible. However, in the common case where a government holds more liabilities than assets, the $r^d - g$ differential is typically the dominant factor. And despite recent increases in market interest rates, $r^d - g$ differentials remain negative in most OECD countries at the starting point for the long-run projections (Figure 5). This reflects the fact that r^d is an average of past interest rates weighted by outstanding debt holdings and adjusts only gradually as debt is rolled over. The $r^a - g$ differentials are similarly negative at the starting point in all countries (Figure 5).

Figure 5. The $r - g$ differentials are generally negative in 2027



Note: $r^d - g$ is the difference between the implicit average nominal interest rate on government financial liabilities and nominal potential GDP growth. $r^a - g$ is the difference between the implicit average nominal interest rate on government financial assets and nominal potential GDP growth. The latter is not shown for Israel as it is a large outlier, but Israel has few government financial assets.
Source: OECD Economic Outlook 119 long-term scenarios database.

The interest rate – growth differential

In the LTM, the short-term real interest rate converges to a neutral rate, assumed to equal trend economic growth, while inflation stabilises at an equilibrium rate embedded in nominal interest rates.[2] Consequently, short-term nominal rates converge to nominal potential growth on a country-specific basis, except in euro area countries, where they converge to euro area-wide potential growth. Long-term nominal interest rates (10-year government bond yields) are modelled as forward convolutions of projected short-term rates, plus a uniform 70 basis point term premium[3] and a country-specific fiscal risk premium based on fiscal positions and other structural factors.[4] The modelling assumptions imply that both short and long-term market interest rates react one-for-one to changes in potential growth rates, all else equal, after some adjustment period.

Implicit average interest rate on debt

The projection of the implicit average interest rate on government debt (r^d) reflects the gradual refinancing of outstanding debt at projected market rates. Newly issued debt is assumed to carry an interest rate equal to one quarter of the short-term rate and three quarters of the long-term rate, implying an effective average maturity of roughly 7½ years. The share of debt maturing each year is taken from country-specific market

[2] The sole price index in the LTM is the GDP deflator, which includes an adjustment for the Harrod-Balassa-Samuelson effect. In advanced countries, this effect is small and equilibrium GDP inflation is close to the consumer price inflation target.

[3] 70 basis points roughly corresponds to the average difference between the long (10-year) and short (3-month) interest rates on government bonds observed over the past two decades in a selection of countries where fiscal risk can be considered to have been negligible (United States, United Kingdom, Canada, Norway, Switzerland, Japan), so as to measure a 'pure' term premium.

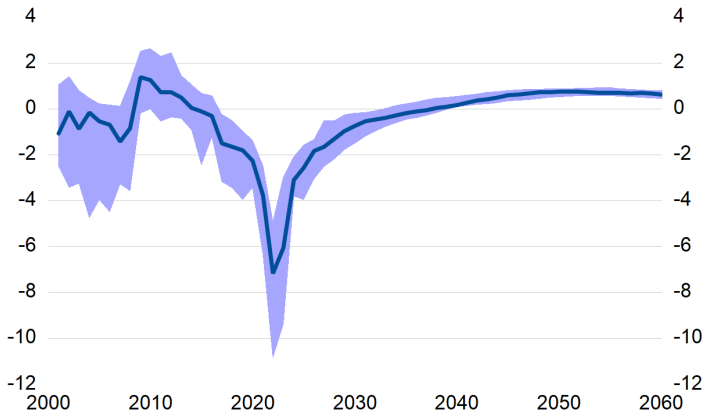
[4] See section 5 of Guillemette (2019[26]) for methodological details on interest rate projections.

data for the initial year but gradually converges to a uniform 20%, approximately the current OECD average.

Because long-term rates exceed short-term rates, they exceed nominal potential growth and $r^d - g$ differentials converge to small positive values over time (Figures 6 and 7). The average $r^d - g$ differential over the projection period is 0.3 pp, broadly aligned with the empirical finding of Wyplosz (2019[13]), who reports an average of 0.1 pp for 22 OECD countries since 1961. By 2060, projected $r^d - g$ differentials are above 1 pp in only a few euro area countries — those with particularly low potential growth relative to the euro area aggregate to which the euro area's neutral interest rate is tied (Croatia, Italy, Lithuania and Latvia) (Figure 7).

Figure 6. The $r^d - g$ differential crucial to debt dynamics normalises in the projection period

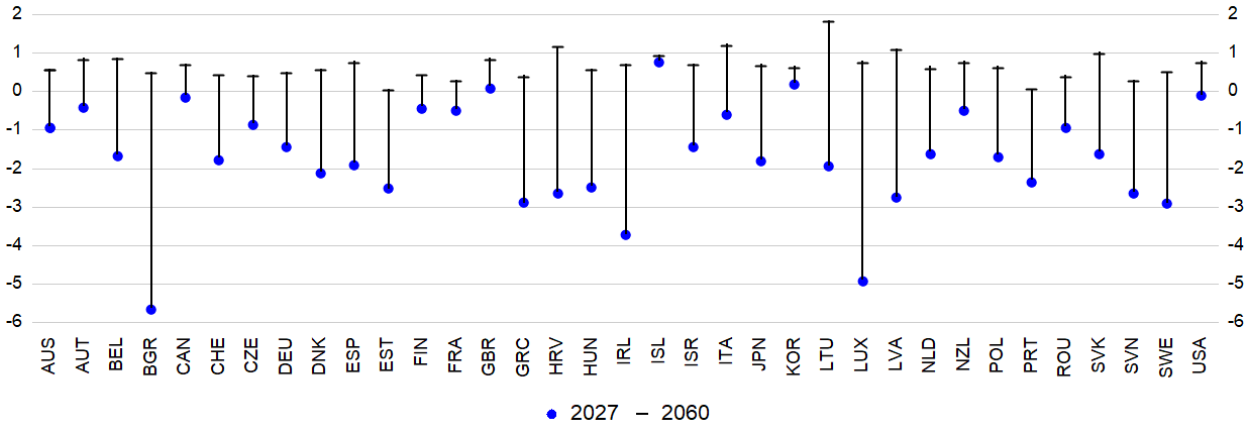
% pts, median and interquartile range



Note: The countries included in the calculation are the same as in Figures 1 to 4. The shaded area is the interquartile range.
Source: OECD Economic Outlook 119 long-term scenarios database.

Figure 7. The $r^d - g$ differentials are all positive by 2060

% pts



Note: $r^d - g$ is the difference between the implicit average nominal interest rate on government financial liabilities and nominal potential GDP growth.
Source: OECD Economic Outlook 119 long-term scenarios database.

The implicit average interest rate on government financial assets

The equivalent differential on assets, $r^a - g$, plays a less important role than $r^d - g$ in fiscal projections because most countries hold lower stocks of financial assets than debt. Changes in $r^a - g$ affect long-run fiscal gaps in the opposite direction to changes in $r^d - g$. An increase in $r^a - g$ implies that a larger share of the financial asset accumulation required to keep assets growing in line with GDP (as assumed in the LTM) comes from interest receipts.

The projection rule for r_t^a has been revised and is now based on a simple error-correction model estimated on a panel of 36 countries (those with a fiscal block in the LTM) using annual data (Table 1). Experimentation with various specifications showed that the domestic short-term interest rate ($IRS_{i,t}$) was the only robust explanatory variable in both the short and long run. Omitting the small, statistically insignificant constant, the rule makes the implicit average rate of return on financial assets converge in the long run to approximately 60% of the short-term interest rate. Because the three-month rate is assumed to equal trend nominal growth in equilibrium, the rule implies that raising potential growth also modestly raises returns on government financial assets. As with $r^d - g$, the $r^a - g$ differentials generally increase over time, but in contrast remain at or below zero in all countries by 2060 (Figure 8).

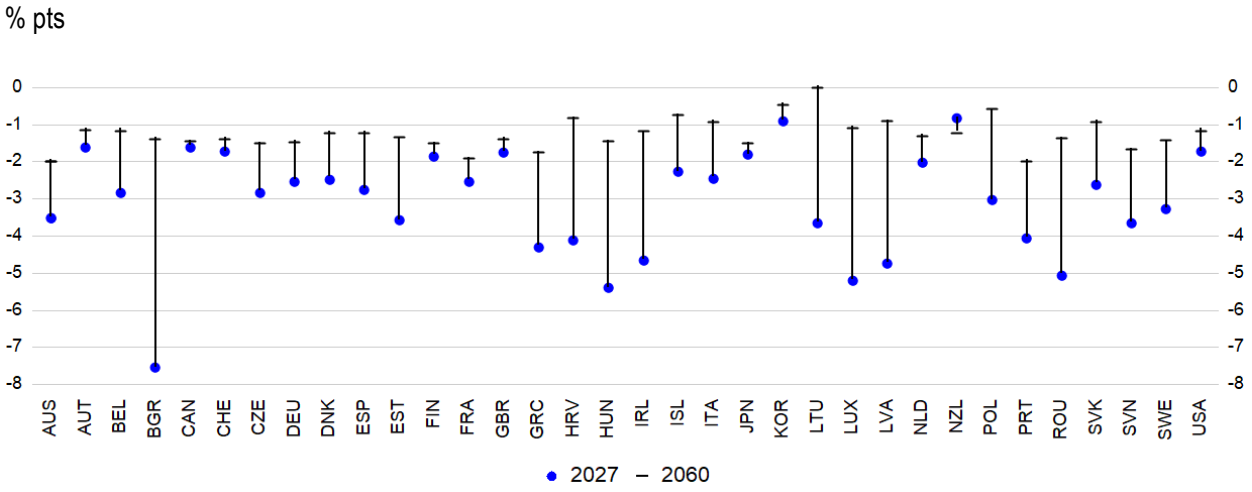
Table 1. Estimated equation for implicit average rate of return on government financial assets

Estimated equation: $\Delta r_{i,t}^a = \beta_1 \cdot \Delta r_{i,t-1}^a + \beta_2 \cdot \Delta IRS_{i,t} + \rho \cdot \left[r_{i,t-1}^a - \alpha - \beta_3 \cdot IRS_{i,t-1}\right] + \varepsilon_{i,t}$

	Estimated coefficient
Short run	
Lagged change in rate of return on assets ($\hat{\beta}_1$)	-0.319***
Change in short-term interest rate ($\hat{\beta}_2$)	0.250**
Long run	
Constant ($\hat{\alpha}$)	-0.035
Short-term interest rate ($\hat{\beta}_3$)	0.617***
Speed of convergence to long run ($\hat{\rho}$)	-0.084**
Estimation period	1993-2024
Number of countries	36
Number of observations	998
Adjusted R^2	0.224

Note: In the specification i indexes countries and t indexes time periods (years). Asterisks (*, **, ***) indicate the statistical significance level (10%, 5%, 1%) of the estimated coefficients based on (White) standard errors robust to arbitrary heteroskedasticity and within-country serial correlation (cross-sectional clustering). All data are from the OECD Economic Outlook 118 database.

Figure 8. The $r^a - g$ differentials remain negative or zero by 2060



Note: $r^a - g$ is the difference between the implicit average nominal interest rate on government financial assets and nominal potential GDP growth. It is not shown for Israel as it is a large outlier, but Israel has very little government financial assets.
Source: OECD Economic Outlook 119 long-term scenarios database.

Accounting for valuation effects on government financial assets

Based on historical OECD experience, the value of government financial assets is assumed to remain stable as a share of GDP over time, consistent with those assets continuing to play a similar role relative to the size of the economy as today. Some countries have explicit objectives to change the assets-to-GDP ratio, for instance to build pension reserves, but this is not currently considered in the LTM.[5]

Maintaining a stable financial-assets-to-GDP ratio requires accumulating assets in line with nominal GDP growth. This accumulation affects the long-run fiscal gap indicator. Previous versions of the LTM assumed that all increases in government financial assets resulted from debt-financed net asset purchases. In practice, however, revaluation effects — mainly unrealised capital gains on equity-type assets — are substantial and contribute a significant share of the required increase in the nominal value of assets to maintain its ratio to GDP. Indeed, excluding net acquisitions of financial assets ($NAFA_{i,t}$), a residual item in the annual change in government financial assets ($\Delta GFA_{i,t}$) can be calculated as:

$$RESID_{i,t} = \frac{\Delta GFA_{i,t} - NAFA_{i,t}}{\Delta GFA_{i,t}} \quad [1]$$

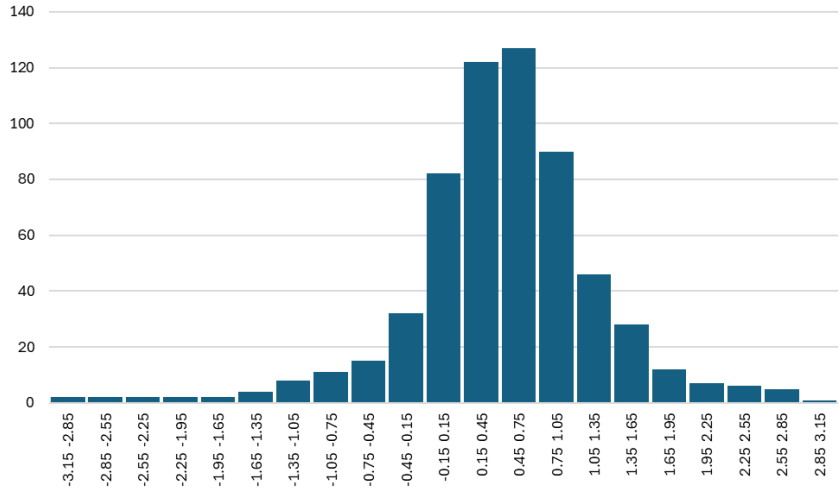
where i indexes the 35 countries (Norway being excluded) and t the years. Over the 652 available observations from 2005 to 2024, the median revaluation effect is 0.51 (Figure 9). On this basis, the LTM

[5] For countries with some pre-funding of future social security obligations (e.g., Canada's CPP/QPP, Japan's GPIF, Korea's NPS, Sweden's AP funds, or the US Social Security Trust Fund), the LTM assumption may depart substantially from national objectives and projections. For instance, in Canada, official projections anticipate significant growth in pension plan (CPP/QPP) assets relative to GDP. If fully reflected on a consolidated general government basis, this accumulation would result in a rising trajectory for financial assets as a share of GDP. More country-specific modelling of pre-funding within social security schemes can be done when the framework is used for tailored analysis, for instance in the context of OECD country surveys.

now adopts the stylised assumption that half of the projected annual increase in government financial assets arises from revaluations.[6]

Figure 9. Share of valuation effects in annual change in government net financial assets, 2005-24

Frequency distribution of $RESID_{i,t}$ in [1]



Note: Some large outliers (< -3 and > 3) are excluded from the chart.
Source: OECD National Accounts (consolidated financial accounts).

The result of the revised projection rule is that the required pace of net asset purchases is lower, reducing the long-run fiscal gap by 0.7 pp of GDP in the median country, and by up to 2-2½ pp in countries with large public financial-asset holdings (e.g. Canada, Finland, Japan).

The overall contribution of changes in $r - g$ to long-run fiscal gaps

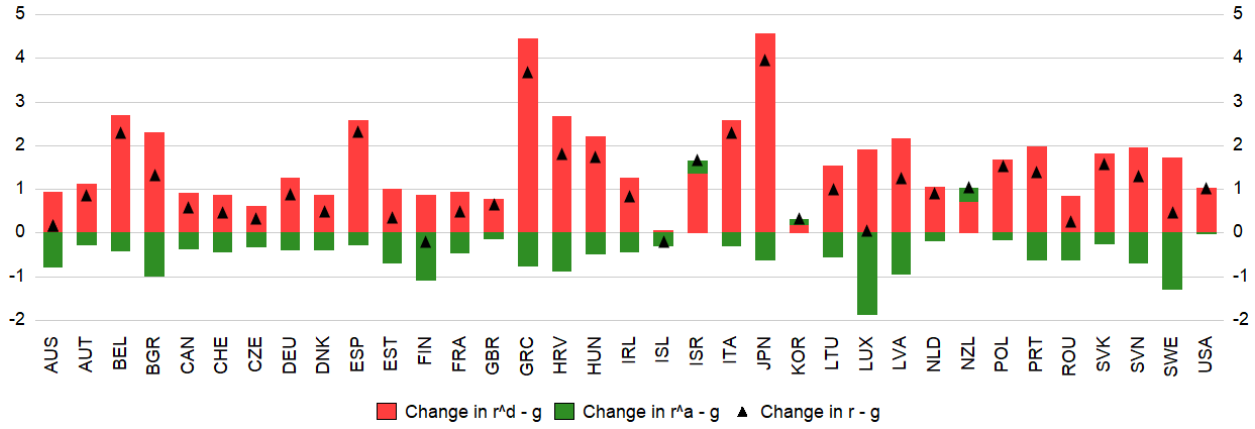
The 'change in $r - g$' component of the long-run fiscal gap summarises the net impact of projected movements in r^d, r^a, and g, the interaction of these variables with stocks of liabilities and assets, and the valuation effect on government financial assets. Projected changes in $r - g$ differentials from 2027 onwards add 0.9 pp of GDP to long-run fiscal gaps in the median country by 2060 (Figures 2 and 10). This can be broken down into a 1.2 pp contribution from the projected change in $r^d - g$ in the median country, which is partly compensated by a 0.4 pp reduction from the projected change in $r^a - g$.[7] These respective effects are much greater in countries with, respectively, high government debt ratios (e.g. Greece, Japan) or a high ratio of government financial assets to GDP (e.g. Finland, Japan).

[6] Revaluation effects also occur for gross government financial liabilities, as these are at market value in the model, but these are not modelled in the LTM. Beyond an initial period of 5-10 years, such revaluations can be expected to be small because the implicit price of bonds moves inversely to interest rates and rates evolve only slowly.

[7] The two median components do not add up to the reported total exactly because medians are not additive.

Figure 10. Decomposition of the 'change in $r - g$' component of the long-run fiscal gap, 2027-60

% pts of potential GDP



Note: The individual contributions are computed using equation [A3] in Appendix A and allowing, respectively, the $(r_t^d - g_t)/(1 + g_t)$ or the $(g_t/2 - r_t^a)/(1 + g_t)$ term to change while holding the other constant.
Source: OECD Economic Outlook 119 long-term scenarios database.

The evolution of primary expenditure

If government primary expenditure were assumed to remain constant as a share of GDP over the projection period, the two previous components would constitute the total fiscal adjustment required to stabilise the debt-to-GDP ratio. However, primary expenditure is generally projected to rise over time due to population ageing and other cost pressures.

Primary expenditure projections are generated in the LTM by linking each spending component to plausible volume and price determinants. These are summarised in Table 2 and explained in detail below. The approach supposes a "business-as-usual" trajectory in which no major additional policy reforms are implemented from 2028, continuing with the approach used previously. Apart from their initial expenditure levels — and the specific exceptions noted below — country-specific features of health, pension, or other programme designs are not modelled explicitly.

Table 2. Summary of baseline assumptions for primary government expenditure

Expenditure component expressed as % of GDP	% of potential GDP in the median country in 2025	Volume determinants	Price assumption
Health and long-term care expenditure	6.6	- Income effect - Ageing effect	GDP deflator + ½ of real wage growth
Pension expenditure	8.4	Proportional to ratio of retirees to employees	GDP deflator + real wage growth
Defence expenditure	2.1	NATO commitment or return to historical long-run average	
Other current primary expenditure	22.2	See Table 3	
Net capital expenditure ex-defence	1.1	Converges to 0.8% of potential GDP	

Note: In the LTM, real wage growth is equal to labour productivity growth by construction.

One important feature of long-run government expenditure projections is the overall degree of pass-through from economy-wide real wage growth into spending levels, which reflects the different degrees of wage indexation assumed for the various spending components. Zero pass-through (purely price indexing) would imply that public sector wages, benefits and social transfers would not keep pace with trend growth in private-sector wages, and thus that public sector workers and benefit recipients would tend to become relatively poorer over time. Full pass-through would imply keeping perfect pace with the private sector but government spending is not composed only of wages and transfers, it also consists of goods and services purchased on the private market which should naturally follow a price as opposed to a wage index. The degree of indexation for overall government spending should thus intuitively fall somewhere between zero and one.

For defence, net capital expenditure and parts of other current primary expenditure, the volume and price assumptions cannot be separated, because LTM projection rules are expressed directly in percentage of GDP terms. However, assuming that projected changes in these expenditure categories are driven by volume rather than relative price effects and weighing the other categories by their relative GDP shares (column 2 of Table 2), then the overall degree of real wage indexation for total primary expenditure in the LTM works out to approximately 54% in the median country. By design, this is approximately the share of expenditure attributable to compensation of employees and cash social transfers in the median OECD country.

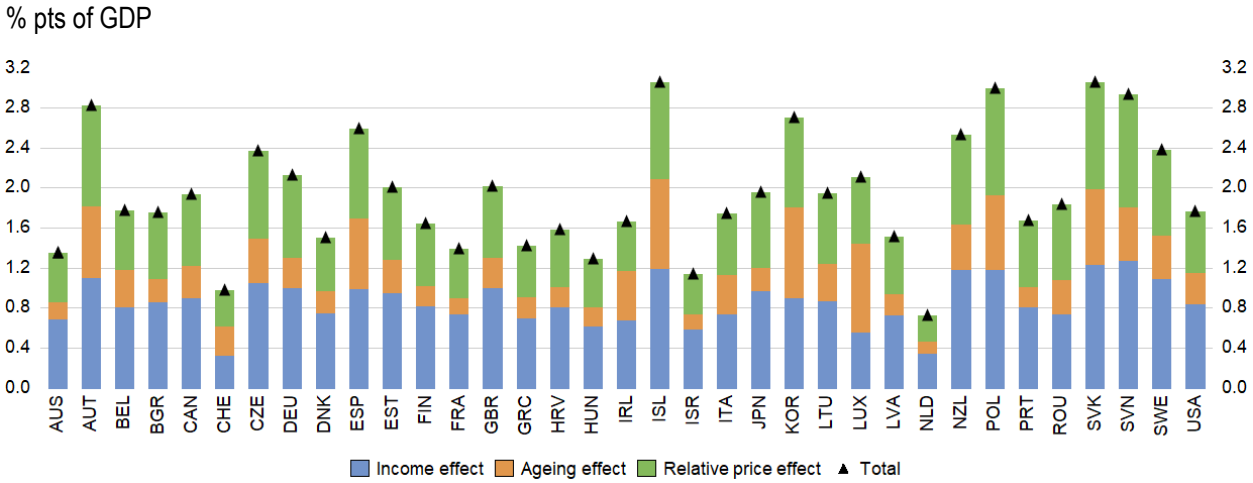
Health and long-term care expenditure

In the LTM, growth in real per capita public expenditure on health and long-term care (measured relative to the GDP deflator) is based on the work of Farag et al. (2012[14]) and Lorenzoni et al. (2019[15]) and driven by three factors: real GDP per capita growth (the income effect), changes in the share of the population aged 65 and over (the ageing effect) and the excess of health-care inflation over general inflation (the Baumol effect).[8] The first two factors primarily influence the volume of services, while the Baumol effect influences their relative price, reflecting the cost pressures typical of labour-intensive sectors where productivity grows more slowly than in the wider economy. The relative price assumption is that, in addition to the general price level (GDP deflator), health and long-term care prices are assumed to increase by one half of economy-wide real wage growth, which is equal to economy-wide labour productivity growth by construction. Given that compensation of employees accounts for 30% to 50% of health spending in OECD countries once accounting for wages embedded in purchased services (intermediate consumption), the assumption implies that wages in this sector rise in line with economy-wide wages. It also implies that faster productivity growth magnifies the Baumol effect.

Under a business-as-usual assumption in which no major policy reforms are implemented, real per-capita public expenditure on health and long-term care is projected to rise by 1.8 pp of GDP between 2027 and 2060 in the median country (Figure 11). Most of this increase is driven by income growth; roughly one-third reflects the Baumol effect; and slightly less than one-fifth arises from population ageing. The ageing effect varies widely across countries depending on demographic trends — for example, it is nearly three times larger in Korea than in Denmark. The Baumol effect is stronger in countries that have lower per capita incomes today because of their higher projected productivity growth through 2060.

[8] See section 4 in Guillemette (2019[26]) for additional methodological details.

Figure 11. Projected change in health and long-term care expenditure, 2027-60



Source: OECD Economic Outlook 119 long-term scenarios database.

Recent evidence suggests that a shift in the nature of technological change could be increasing the rate of cost-saving innovation in health care (Smith, Newhouse and Cuckler, 2022[16]; Cutler and Klarnet, 2026[17]). In light of this evidence, growth in real per capita health spending over the 2014-24 period was decomposed on the basis of the LTM equation to assess its recent performance. For the median country, roughly one third of the nearly 25% increase in spending is explained by the income effect, with the ageing and Baumol effects each accounting for about one quarter, leaving a small positive residual. In other words, the equation slightly under-predicted health spending growth in most, though not all, countries over this ten-year period. Nevertheless, future cost-saving innovations could dampen health spending growth, posing a downside risk to the LTM's projections.

Pension expenditure

Public pension expenditure (defined as cash expenditure on old-age and survivors' pensions) is projected in the LTM based on the change in the ratio of retirees to workers and the evolution of the average benefit ratio (the ratio of the average public pension benefit to the average wage). It is projected to increase by 2.2 pp of GDP in the median country between 2027 and 2060 (Figure 2, Panel A). Projected increases in public pension expenditure tend to be smaller in countries that have legislated increases in statutory retirement ages, especially those linking future retirement ages to life expectancy (e.g. Finland), and larger in countries with low statutory retirement ages and particularly unfavourable demographic profiles (e.g. Poland).

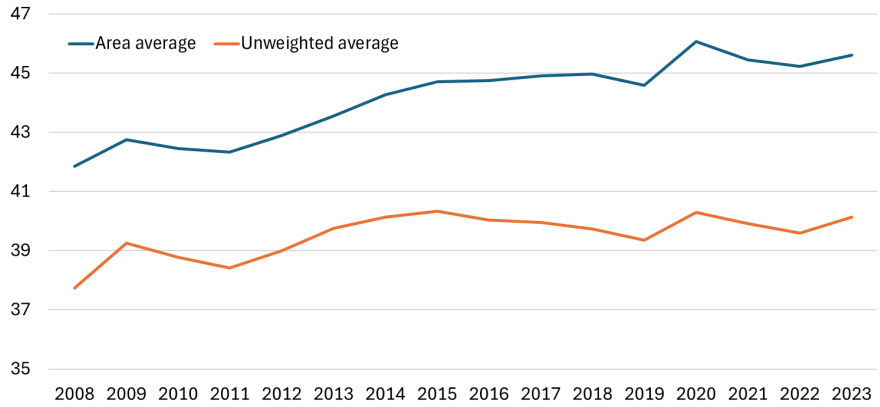
In the business-as-usual scenario, country-specific average pension benefit ratios are assumed to remain constant over the projection period, with pensions rising in line with the average wage. However, in some countries, existing pension legislation implies a substantial reduction in benefit ratios over time, due to indexation to price inflation (e.g. France), or to a mixed index of price and wage inflation (e.g. Finland, Switzerland) or through other changes to pension entitlements such as those for new entrants (e.g. Italy). As a result, the LTM may overstate the increase in pension expenditure if these policies remain unchanged. The LTM approach also differs markedly from the European Commission's central scenario, which reflects legislated pension system parameters, including affordability rules and other mechanisms that, if unchanged, imply an 8.4 pp decline in the average benefit ratio at the EU level between 2025 and 2060 (European Commission, 2024[18]).

The constant-benefit-ratio assumption in the LTM is motivated by two main considerations:

- *Simplicity*. It can be applied uniformly across EU and non-EU countries, whereas setting country-specific assumptions for non-EU countries would be difficult without comparable official projections.
- *Plausibility*. A continuous decline in average pension benefit ratios over several decades, as less than full wage indexation would imply, seems unlikely given the salience of inequality and old-age poverty and the growing political weight of older voters. In practice, governments often introduce ad hoc adjustments to maintain pensioners' relative purchasing power. This pattern is reflected in historical outcomes. For example, the 2015 *Ageing Report* projected a 1.9 pp decline between 2013 and 2025 (European Commission, 2015[19]), whereas the EU average benefit ratio has risen since 2013, as well as in years prior (Figure 12). This suggests an effective link between wages and average pension benefits. Assuming this link persists provides a more realistic foundation for long-run fiscal analysis, while alternative policy choices regarding pension generosity can be explored in scenario variants (see section 3.5).

Figure 12. Average pension benefit ratio in the European Union

Average old-age and survivor pension, % of average full time-adjusted salary



Source: Eurostat and author's calculations.

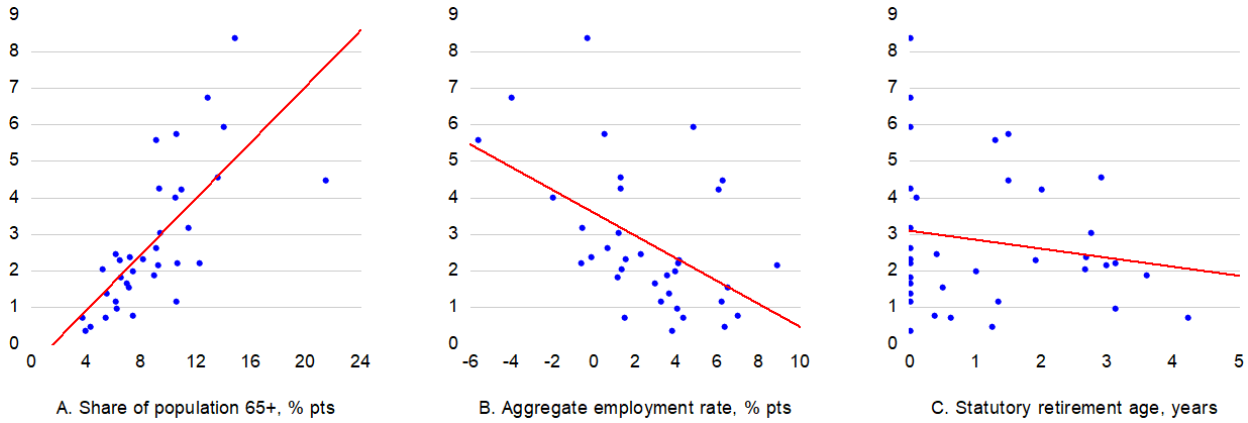
Given the assumption of constant average benefit ratios, the LTM projection reflects only the change in the ratio of retirees to workers, driven by evolving demographics and projected employment dynamics. While an exact decomposition is not possible, three factors are central:

- *The speed of population ageing*. Faster ageing leads to a larger increase in the retirees-to-workers ratio and, consequently, a larger rise in pension expenditure as a share of GDP (Figure 13, Panel A). The LTM's demographic projections are based on the United Nations Population Division's (UNPD) *World Population Prospects* (2024 revision, medium scenario).
- *Employment rates across age groups and their evolution*. Employment rates are projected using a cohort-based model (Box 1). Retirement rates depend on exit rates from employment across the 50-54 to 75+ age ranges. Differences in employment rates across age groups therefore affect the number of retirees, while changes in the aggregate employment rate influence the denominator of the retirees-to-workers ratio. Generally, higher projected increases in aggregate employment reduce projected pension expenditure as a share of GDP (Figure 13, Panel B).
- *Already-legislated changes in statutory retirement ages*. Projected employment rates incorporate legislated changes in statutory retirement ages. Once legislated, these tend to be more permanent than benefit-indexation mechanisms, which justifies including them in the baseline scenario.

Changes to statutory retirement ages do not translate one-for-one into changes to effective retirement ages. Instead, a one-year increase in the statutory retirement age raises employment rates in the 55-74 age group by 1.6 pp in countries with significant voluntary occupational pensions and by 2.2 pp in countries where such plans are less prevalent, based on Morgavi (2024[20]). Expressed differently, a one-year increase raises the average effective retirement age by 3-5 months in the former group and 4-7 months in the latter.[9] These changes reduce the number of retirees and increase the number of workers, thereby lowering pension expenditure as a share of GDP (Figure 13, Panel C).

Figure 13. Three broad factors affecting pension expenditure projections

The y-axes of all three panels show the projected change in pension expenditure between 2027 and 2060 in % pts of potential GDP and the x-axes show the change in the indicated variables over the same period



Note: The x-axis of Panel C is based on both the male and female statutory retirement ages, with weights of one-half each.
Source: OECD Economic Outlook 119 long-term scenarios database.

Box 1. The cohort employment model and the calculation of average effective retirement ages

The LTM includes a cohort-based employment model in which the evolution of trend employment ($ETPT$) is driven by three main dynamics: changes in the size of the working-age population (ages 15 to 74), shifts in its age composition, and trends in the employment rates of different age-sex groups. The size and composition of the working-age population are taken from the United Nations Population Division's (UNPD) *World Population Prospects* (2024 revision, medium variant) and are exogenous to the model. Trends in age-sex-specific employment rates are projected using a cohort approach applied to historical data (Cavalleri and Guillemette, 2017[21]). These generational trends capture societal changes, such as rising female labour-force participation, as well as structural changes like increased educational attainment.

For example, in many countries, younger women exhibit much higher employment rates today than in the past. By holding the most recently estimated entry and exit rates constant over the projection horizon, the cohort model typically projects that employment rates for older women will rise over time as today's younger cohorts move through their life cycle.

[9] These impacts differ between countries because the calculation of average effective retirement ages depends on population weights for different age/sex groups.

Already-legislated changes in statutory retirement ages are incorporated into the projected employment rates for individuals aged 55 and over.

The average effective retirement ages for sex $S \in \{M, F\}$ at time t (ARA_t^S) is calculated using exit rates from employment across the age groups 50-54 and above. For example, the difference between the employment rate of the 55-59 age group at time t ($ER_t^{S,5559}$) and that of the 50-54 age group five years earlier ($ER_{t-5}^{S,5054}$) gives the exit rate for that cohort. The average retirement age for that group of retirees is set at the mid-point between the two age groups (54.5). The overall average effective retirement age is computed in that manner using all age groups and weighting by employment:

The overall average effective retirement age is computed in that manner using all age groups and weighting by employment:

$$ARA_t^S = \frac{54.5 \cdot \left(ER_{t-5}^{S,5054} - ER_t^{S,5559}\right) \cdot POP_{t-5}^{S,5054} + \cdots + 74.5 \cdot \left(ER_{t-5}^{S,7074} - 0\right) \cdot POP_{t-5}^{S,7074}}{\left(ER_{t-5}^{S,5054} - ER_t^{S,5559}\right) \cdot POP_{t-5}^{S,5054} + \cdots + \left(ER_{t-5}^{S,7074} - 0\right) \cdot POP_{t-5}^{S,7074}}$$

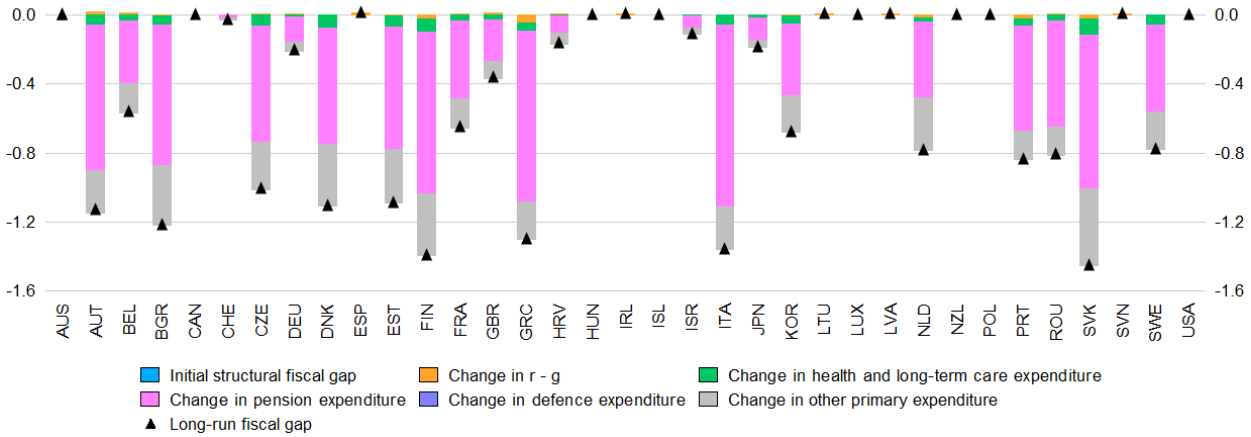
where $POP_t^{S,A}$ is the population of sex S in age group A.

A constant-benefit ratio scenario was also presented in the European Commission's 2024 *Ageing Report*. Focusing on the euro area over the 2025-2060 period, the Commission projected an increase in pension expenditure of about 1½ pp of GDP, whereas the LTM projects a larger rise of 2½ pp. Simulations using alternative demographic assumptions (see section 3.4) suggest that most of this difference reflects variations in the underlying demographic projections. For example, the old-age population dependency ratio for the EU area — defined as the population aged 65 and over relative to those aged 20-64 — increases by 20 pp between 2025 and 2060 in the Eurostat population projections used in the Commission's assessment. In contrast, this ratio increases by 23.8 pp in the UNPD projections used in the LTM. A key factor behind this difference is that Eurostat typically assumes higher net international migration over the coming decades than the UNPD.

To illustrate more clearly the fiscal benefits of legislated pension reforms, a scenario is constructed in which all countries maintain statutory retirement ages at their 2027 levels. The contribution of legislated reforms to reducing long-run fiscal gaps is then measured by comparing this scenario with the baseline. Linking statutory retirement ages to life expectancy substantially reduces long-run fiscal gaps in Bulgaria, Denmark, Estonia, Finland, Greece, Italy, the Netherlands, Portugal, the Slovak Republic and Sweden, by 1.2 pp in the median country (Figure 14). Even without an explicit link to life expectancy, countries that have planned significant reforms, such as Austria, the Czech Republic and Romania, also benefit. The reduction in long-run fiscal gaps emerges not only via lower pension expenditure, but also through the 'other primary expenditure' component, reflecting an improved employment-to-population ratio relative to baseline (as explained in section 2.5.4).

Figure 14. The contribution of already-legislated pension age reforms to reducing long-run fiscal gaps

Difference from baseline in % pts of potential GDP



Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

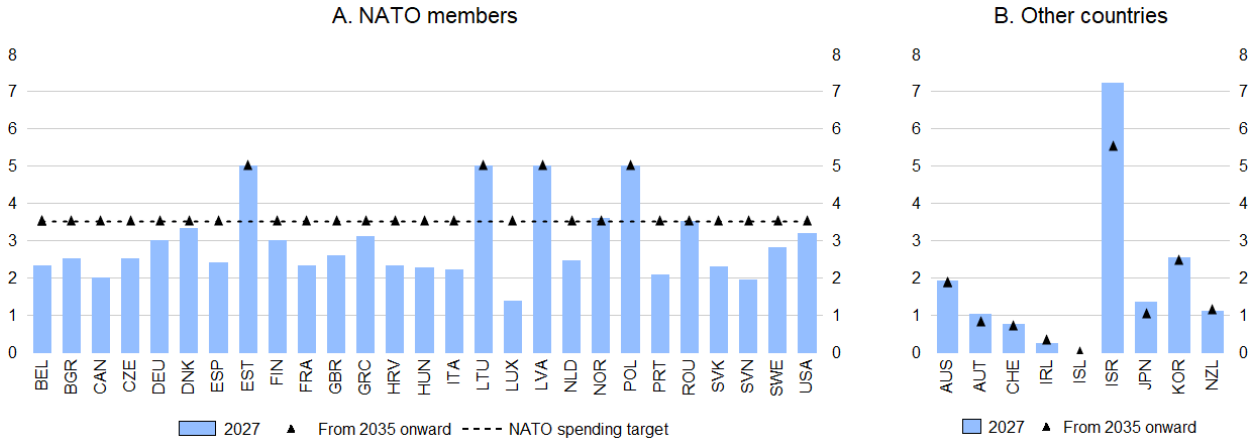
Defence expenditure

Defence expenditure is incorporated explicitly in the long-run fiscal gap indicator for the first time in this paper. The starting point for the projections is each country's defence spending as a percentage of GDP, based on data from the Stockholm International Peace Research Institute (SIPRI) (Figure 15). For the projection period, assumptions differ between North Atlantic Treaty Organization (NATO) members and non-members:

- *NATO members*: Defence expenditure is assumed to:
  - Rise to 5% of GDP by 2027 and remain at this level thereafter in Estonia, Latvia, Lithuania and Poland. These members have already announced a near-term target of 5% of GDP and, given their geography, it seems reasonable to assume that this spending level will be maintained over the long run. Since defence spending is expected to be at the target in 2027, this assumption implies that defence does not contribute to the long-run fiscal gaps in these countries by construction.
  - Rise to 3.5% of GDP by 2035 and remain at this level thereafter in other NATO countries. This target is based on the 2025 NATO Summit in The Hague, where Members committed to investing at least 3.5% of GDP annually on core defence requirements and on meeting NATO Capability Targets (Conigrave and Shin, 2026[22]). Iceland is an exception as it has no military and is assumed not to increase spending. The assumption implies a 1.1 pp of GDP contribution to the long-run fiscal gap of the median remaining country by 2035.
- *Non-NATO members*: Defence expenditure is assumed to return to its 15-year historical average (2010-2025) by 2035. This implies limited changes from the latest SIPRI estimate (2025) for most countries, except Israel, where exceptionally high spending in 2025 leads to a projected decline.

Figure 15. Defence expenditure

% of potential GDP



Source: Stockholm International Peace Research Institute (SIPRI) military expenditure database for historical data.

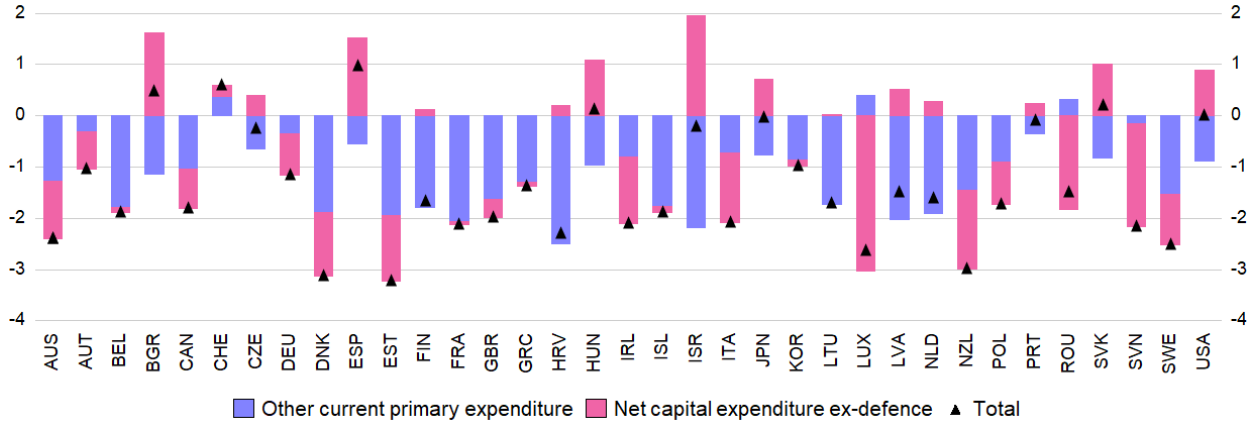
Defence expenditure is added as a refinement of the long-run fiscal gap indicator. For now, it is not assumed to have any longer-run effects on potential output, though there are some potential channels through which it could do so, including defence R&D and (some) capital accumulation (OECD, 2026[23]; Conigrave and Shin, 2026[22]).

Other primary expenditure

Other primary expenditure refers to all primary spending not covered previously (health, long-term care, pensions, and defence). In the median country, it is projected to fall by 1.7 pp of GDP between 2027 and 2060 given the approach set out below (Figure 16). Other primary expenditure can be split between other current primary expenditure and net capital expenditure on non-defence items, since defence expenditure includes defence-related capital expenditures.

Figure 16. Change in other primary expenditure, 2027-60

% pts of potential GDP



Source: OECD Economic Outlook 119 long-term scenarios database.

Other current primary expenditure

The approach to projecting other current primary expenditure has been refined from previous versions of the LTM to better reflect the composition of other primary government expenditure in the average OECD country. The previous approach was based on two simplifying assumptions regarding the volume and relative price of government services in this expenditure item:[10]

- Volume component: Real per capita spending in this category was assumed to remain constant over time. Accordingly, the volume of government services was projected to grow in line with total population.
- Price component: The price of government services was assumed to increase at the rate of nominal wages.

Under these assumptions, changes in the ratio of other current primary expenditure to GDP depended solely on the population-to-employment ratio. As a result, productivity-enhancing reforms had no long-run impact on this spending category as a share of GDP.

In practice, while full wage indexation is appropriate for major components such as compensation of employees and social transfers, other items are more appropriately linked to the general price level. These include parts of intermediate consumption and purchased market output (e.g. social transfers in kind) that reflect competitively supplied goods and services (such as electricity, fuel, and office equipment), as well as subsidies. In these areas, productivity gains can cause spending to decline as a share of GDP. Indexing these items to prices rather than wages implies that their share of GDP declines as economy-wide productivity increases.

The revised approach is this paper distinguishes four expenditure categories based on COFOG data, each with specific assumptions regarding volume and price developments (Table 3):

1. Education (22% of other current primary expenditure): The volume component is assumed to evolve in proportion to the population aged 5 to 24. The price component is assumed to increase with the GDP deflator plus two-thirds of real wage growth. This implies a gradual rise in the relative price of education services, reflecting a Baumol-type effect similar to that assumed for health and long-term care, but with a higher coefficient (two-thirds rather than one-half) to reflect this sector's higher labour intensity. This allows salaries in the education sector to rise in line with economy-wide wages.
2. Social protection, excluding old-age and survivors' benefits (31%): The volume component is assumed to grow in line with the total population. The price component follows the GDP deflator plus real wage growth, implying that the average benefit ratio (average benefit relative to the average wage) remains constant over time. This ensures consistency with the treatment of public pension expenditure.
3. Housing and community amenities (2%): The volume component is assumed to grow with the total population, while the price component follows the GDP deflator only.
4. Other categories (45%): Expenditure in the remaining categories, including for example general public services and environmental protection, is assumed to remain constant as a share of nominal GDP. For these items, distinguishing between volume and price components is conceptually difficult and a constant-share-of-GDP assumption is consistent with long-term trends in countries with available historical data.

In practice, the LTM does not explicitly disaggregate other current primary expenditure into these sub-categories. Instead, total expenditure growth is projected as a weighted average of the volume and price determinants described above. The shares are based on unweighted OECD averages, assumed to be

[10] See section 3.1.2 in Guillemette and Turner (2017[29]) for more details.

constant over time and uniform across countries. This simplified approach reflects data limitations, as detailed COFOG information is unavailable for some countries and is subject to significant reporting lags.

Table 3. Assumptions for other current primary government expenditure

Component	% of other current primary expenditure in average OECD country	Volume determinant	Price assumption
Education	22	Population aged 5 to 24	GDP deflator + 2/3 real wage growth
Social protection (excluding old-age and survivor benefits)	31	Population	GDP deflator + real wage growth
Housing and community amenities	2		GDP deflator
General public services	15	Constant as a % of nominal GDP	
Public order and safety	7		
Economic affairs	16		
Environmental protection	3		
Recreation, culture and religion	4		

Notes: In the LTM, real wage growth is equal to labour productivity growth by construction. The total current expenditure from which the percentages are calculated exclude health and long-term care, pension and defence expenditure. The percentages are simple unweighted averages across OECD countries for the latest available year.
Source: OECD Annual government expenditure by function (COFOG).

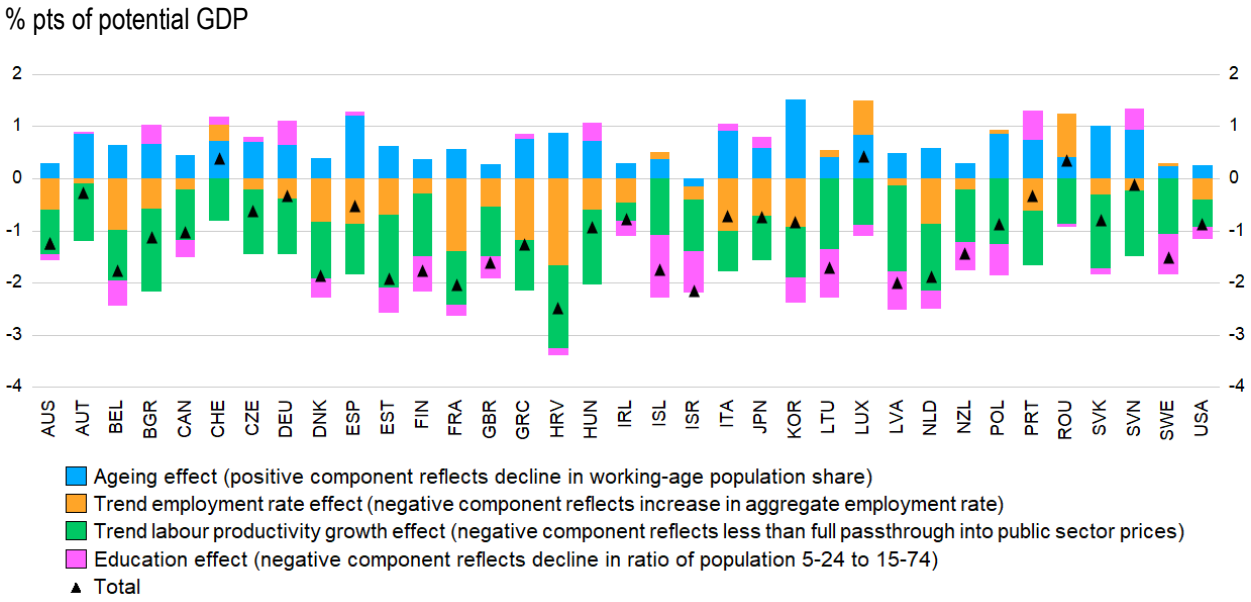
Given the framework and the assumptions just described, other current primary expenditure is generally projected to decline when expressed as a share of GDP, driven primarily by two factors: (*i*) the population-to-employment ratio, because parts of spending are linked to total population, whereas GDP depends on employment, and (*ii*) labour productivity growth, reflecting the varying degrees of pass-through from wages into spending across expenditure components. The change over time can be decomposed more finely into four elements (Figure 17), the first three of which relate to factor *(i)* above:

- Ageing effect: This component captures changes in the share of the population that is of working age. As this share is projected to decline across OECD countries, upward pressure is placed on other current primary expenditure as a share of GDP (blue component in Figure 17, amounting to +0.6 pp in the median country). The effect is particularly strong in Korea, reflecting adverse demographic trends and very low fertility.
- Trend employment rate effect: The aggregate employment rate is projected to rise in most OECD countries, exerting downward pressure on other current primary expenditure as a share of GDP (yellow component in Figure 17, -0.4 pp in the median country). Two opposing forces shape this outcome:
  - A composition effect, as employment rates are typically lower among older age groups.
  - Continued convergence of female employment rates toward male rates, which occurs mechanically in the LTM's cohort-based framework as the present higher employment rate among younger female cohorts persists over their life cycle (Box 1).

  In most countries, the catch-up in female employment more than offsets the negative composition effect, resulting in an increase in the aggregate employment rate. However, in countries where female employment is already high (e.g. Switzerland) or is projected to decline due to cohort dynamics (e.g. Romania), the aggregate employment rate is expected to remain stable or fall. These countries tend to show flat or rising other current primary expenditure as a share of GDP over the projection period. The evolution of female employment is therefore a key driver of cross-country differences in projected spending outcomes.

- Education effect: This captures changes in the ratio of the population aged 5 to 24 (which determines the volume of education spending) to the working-age population. Owing to persistently low fertility rates, this component is generally negative (fuchsia component in Figure 17, -0.2 pp in the median country).
- Trend labour productivity growth effect (part of factor *ii*): This component is negative for all countries and reflects the revised modelling approach, under which part of other current primary expenditure is no longer fully indexed to real wage growth. As a result, the projected increase in this spending category as a share of GDP is lower than in earlier model versions, reflecting governments' gains from declining relative prices of certain purchased goods and services (green component in Figure 17, -1.0 pp in the median country).

Figure 17. Change in other current primary expenditure as a share of GDP between 2027 and 2060



Source: OECD Economic Outlook 119 long-term scenarios database.

Altogether, other current primary expenditure is projected to decline by 1 pp of GDP between 2027 and 2060 in the median country. Although this downward trend may appear counterintuitive, it is broadly consistent with historical developments. Among the 17 countries for which historical data are available, other current primary expenditure declined by 0.6 pp of GDP in the median country between 1995 and 2025. This comparison should be interpreted with caution, however. The historical series are highly cyclical and, more importantly, the projection assumption of no major policy change did not hold in the past. A meaningful comparison would require isolating and removing the impact of policy changes from historical data, which is inherently difficult. Over the period to 2060, the expected decline in other current primary expenditure as a share of GDP is driven primarily by projected increases in aggregate employment rates and labour productivity. If these improvements fail to materialise, or are weaker than anticipated, additional fiscal pressures will arise.

Net capital expenditure ex-defence

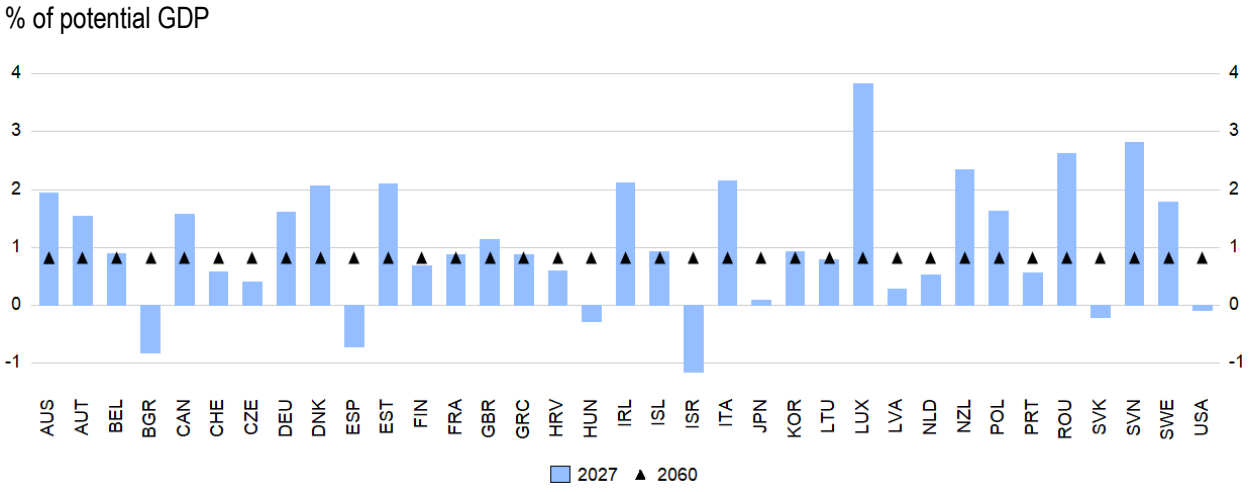
Net capital expenditure is defined as government gross fixed capital formation plus net capital transfers minus consumption of fixed capital. Historical COFOG data are used to exclude defence-related net capital expenditure, which is included in defence spending. When COFOG data are unavailable, total net capital

expenditure is reduced by 25%, approximating the average defence share observed in countries with COFOG data. For convenience, the projection approach focuses on the aggregate net concept to avoid having to project components (capital transfers and consumption of fixed capital) separately.

This component is challenging to project because its long-term drivers are less obvious than for health or pension spending and the historical data are not adjusted for the cycle and highly volatile. Capital transfers often reflect one-off items, such as radio spectrum sales or special EU funds (e.g. NextGenerationEU), making projections based on initial values undesirable. It is assumed that net capital expenditure ex-defence gradually converges to 0.8% of GDP in all countries, which is slightly below the 1995-2025 median of 1.1%. However, it is consistent with the declining path for total investment in the LTM baseline, which reflects the model's targeting a fixed long-run capital-output ratio and slowing potential growth.

The projection rule implies rising net capital expenditure ex-defence in countries where initial-year spending is lower than 0.8% of potential GDP (e.g. Bulgaria), and the opposite where it is higher (Figure 18). The substantial dispersion across countries in the government share of total net investment gradually compresses into a 6-9% range, in line with historical experience (Figure 19). In the median country, the contribution of this component to the long-run fiscal gap is a marginal -0.1 pp (Figure 16). However, the simple projection rule does not account for increasing investment needs, including those related to the energy transition, climate-change mitigation and adaptation, and other long-term priorities. One of these additional pressures, the need for investment in renewable electricity generation, is discussed in the next section.

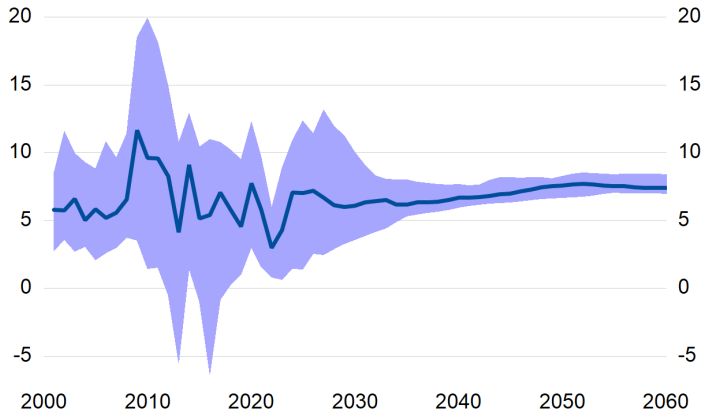
Figure 18. Net capital expenditure ex-defence is assumed to converge to 0.8% of potential GDP



Source: OECD Economic Outlook 119 long-term scenarios database.

Figure 19. Government net capital expenditure ex-defence as a share of total net investment

Per cent, median and interquartile range



Note: The shaded area is the interquartile range.
Source: OECD Economic Outlook 119 long-term scenarios database.

3. Some sensitivity analyses

This section looks at the sensitivity of the long-run fiscal gap estimates to some of the underlying assumptions and trends.

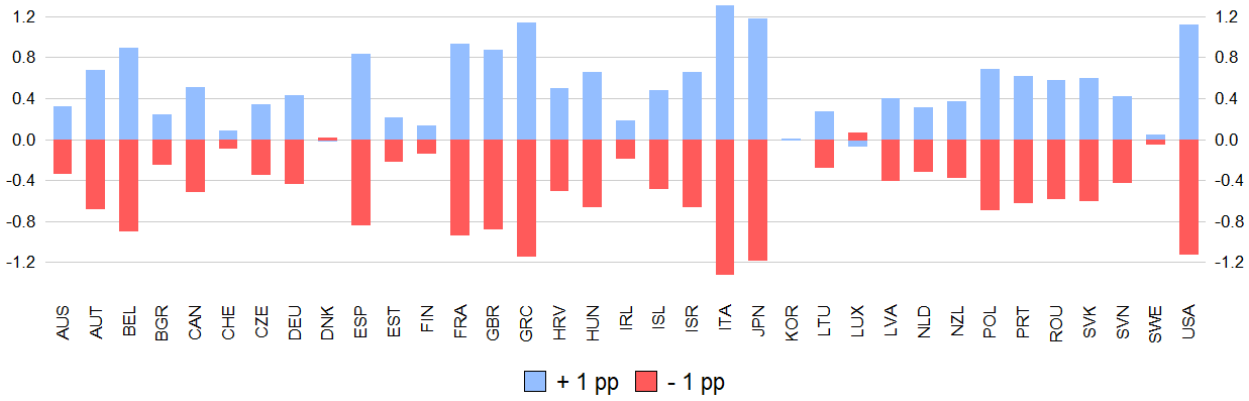
Sensitivity to interest rates

Given the substantial uncertainty surrounding the future path of interest rates, two alternative scenarios assess the impact of lower and higher interest rate trajectories relative to the baseline. These interest rate shocks are introduced by adding one percentage point (positive-risk scenario) or subtracting one percentage point (negative-risk scenario) from the equilibrium short-term neutral interest rate in all countries. The shocks feed quickly into short- and long-term market interest rates, and more gradually into implicit average interest rates on government debt (depending on country-specific debt maturity structures) and on the rate of return on government financial assets, as previously explained.

Higher interest rates than in the baseline scenario increase the long-run fiscal gap, while lower rates reduce it (Figure 20). Unsurprisingly, a country's gross government debt position is the dominant factor shaping its sensitivity to interest rate shocks. In the higher-interest-rate scenario, long-run fiscal gaps increase by as much as 1-1¼ pp of GDP in countries with the largest debt burdens, such as Greece, Italy and Japan. The pattern is reversed in the lower-interest-rate scenario, although the same countries remain the most affected.

Figure 20. Sensitivity of long-run fiscal gap to +/- 1 percentage point on interest rates

Difference from baseline, % pts of potential GDP



Note: Following the decomposition in Figure 2, Panel A, the change in long-run fiscal gaps relative to baseline occurs exclusively in the 'change in $r - g$' component.
Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

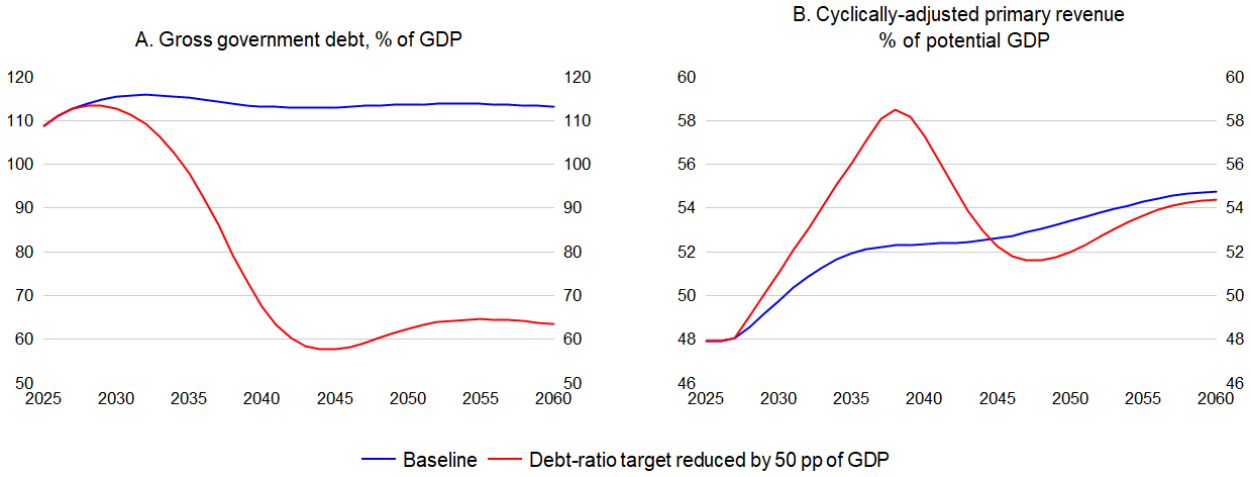
Sensitivity to the gross debt-to-GDP ratio target

Because a wide range of government debt ratios is potentially compatible with long-run fiscal sustainability, one advantage of the LTM's long-run fiscal gap indicator is that it is not very sensitive to the debt-ratio target. The required fiscal adjustment between 2027 and 2060 depends mainly on initial fiscal positions and projected expenditure trends. However, the evolution of the revenue instrument in the intervening years can be sensitive to the debt-ratio target.

The indicator's sensitivity to the debt target can be illustrated using the example of Belgium, whose government debt ratio is close to the OECD average of 110% of GDP. In a "rebuilding fiscal buffers" scenario, Belgium is assumed to gradually reduce its debt-to-GDP ratio by 50 pp rather than stabilising it at the initial level (Figure 21, Panel A). Once the lower debt target is reached, lower interest payments imply a commensurate decline in the level of structural primary revenue, the model's technical instrument for debt stabilisation. The impact is modest: around ½ pp of GDP (Figure 21, Panel B).[11]

[11] Referring to equation [A3] in Appendix A and ignoring the impact of a lower debt ratio on the interest rate via the fiscal risk premium (which is taken into account in the LTM), the impact is approximately equal to $-50 \cdot \left(r^{d}_{2060} - g_{2060}\right)/(1 + g_{2060}) = -0.44$.

Figure 21. Impact of lowering gross indebtedness by 50 pp of GDP



Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

While changing the debt target has little effect on the long-run fiscal gap, it strongly influences the interim fiscal trajectory. Reducing the debt ratio requires not only closing the initial structural fiscal gap and offsetting rising expenditure, as in the baseline, but also sustaining a sufficiently strong fiscal balance to reach the new debt target. Multiple paths could achieve this, depending on the timing, feasible annual consolidation pace, and implementation constraints. A front-loaded adjustment produces an early steep improvement in the fiscal balance followed by a decline, while others smooth the adjustment over a longer period at a lower peak.

Regardless of the path, a much larger fiscal adjustment than in the baseline scenario would be needed over an extended interim period. In the Belgium example, the model's revenue instrument is 2.9 pp of GDP above baseline over the 2028-45 period, peaking at 6.2 pp above baseline (Figure 21, Panel B). While the exact numbers and the fiscal instrument used are illustrative, the key takeaway is that achieving large debt reductions would require a substantial fiscal effort over the coming decades in most countries.

Sensitivity to growth rates and the sources of growth

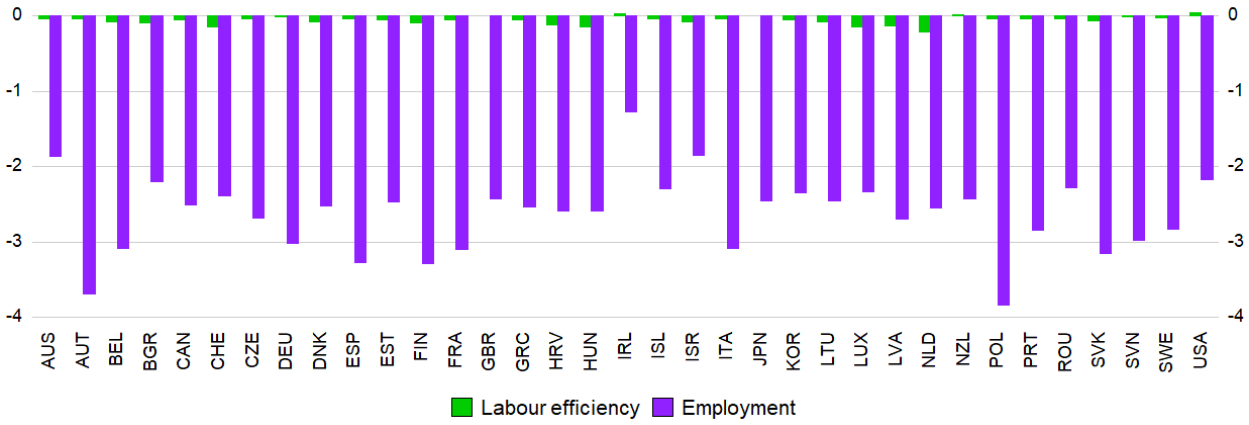
Boosting growth could help narrow fiscal deficits and improve long-run fiscal sustainability, although it is unlikely to eliminate the need for fiscal adjustment entirely. To assess the sensitivity of long-run fiscal gap estimates to stronger growth, two scenarios examine the effects of successful reforms that entail no direct fiscal costs, thereby isolating the pure impact of higher growth. In both scenarios, average annual potential growth rises by around 0.3 percentage points between 2028 and 2060, resulting in a cumulative increase of about 10% in the level of potential output by 2060. In the first scenario, this increase is driven by a 10% rise in trend labour efficiency; in the second, by a 10% increase in employment (with the working-age population unchanged, implying higher employment rates). In both cases, reforms raise potential growth in a hump-shaped pattern relative to baseline, consistent with the treatment of structural reforms in the LTM: potential growth gradually increases before returning to baseline.

The fiscal effects differ markedly across the two scenarios. In the higher labour-efficiency scenario, rising incomes increase health and long-term care spending by around 0.2 pp of GDP by 2060. At the same time, other primary expenditure declines by a broadly similar magnitude as a share of GDP because wage growth is only partially passed through into government spending (see section 2). As a result, the long-run fiscal gap for the median country remains broadly unchanged relative to baseline (Figure 22), although in

some countries the increase in health and long-term care spending slightly outweighs the reduction in other expenditures, leading to a marginal widening of the fiscal gap.

Figure 22. Impact on long-run fiscal gap of a 10% boost to potential output by 2060 according to the supply component responsible

Difference from baseline, % pts of potential GDP



Note: Both scenarios implement a gradual increase in the level of potential output that cumulates to an approximately 10% increase by 2060, the first using higher labour efficiency growth and the second using higher employment rates. Over the simulation period 2028-60, the shocks correspond to an increase in the average annual growth rate of potential output by approximately 0.3 pp.
Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

In the higher-employment scenario, the same income effect places upward pressure on health and long-term care spending. However, this is more than offset by other effects, especially reductions in pension and other primary expenditures as a share of GDP, resulting in substantially smaller long-run fiscal gaps — by 2½ pp in the median country.

This difference reflects a key modelling assumption in the LTM: economy-wide productivity growth raises wages across all sectors, including the public sector. As a result, productivity-enhancing reforms increase the relative price of labour-intensive government services over time, as well as the real value of pension and social protection spending. Consequently, when expenditure is measured relative to GDP, part of the fiscal benefit from higher productivity-driven growth is offset by rising public spending. Employment-driven growth does not generate the same offsetting effect. Although the degree of long-run wage indexation across expenditure categories is uncertain — and alternative assumptions would affect the quantitative results — the qualitative conclusion is robust: different sources of growth do not improve fiscal sustainability to the same extent.

The fiscal gains from stronger growth would be somewhat larger if the LTM made different assumptions:

- Lower than one-to-one long-run pass-through from potential growth to interest rates: As discussed previously, the LTM assumes that market interest rates gradually adjust fully to changes in potential growth, implying that growth shocks affect the long-run fiscal gap primarily through their impact on expenditure ratios. Allowing for less than full pass-through would therefore modestly increase the fiscal benefits of higher growth, although it would not materially alter the difference between the two scenarios.
- Permanent growth impacts: If reforms permanently increased growth rather than temporarily, the reduction in long-run fiscal gaps would be larger.

- Price indexing of tax and benefit thresholds: The LTM implicitly assumes that tax and benefits thresholds are indexed to wages as opposed to prices. If the latter were the case for an extended period, more people would be pushed into higher tax brackets and lower benefit rates, increasing the fiscal gains from productivity growth.
- Relatively higher productivity gains in the public sector: The implications of the productivity scenario could differ substantially if productivity gains were higher in the public sector than in the rest of the economy. For example, education and health systems could potentially use advances in artificial intelligence to raise productivity relative to the private sector. If public-sector wages continued to track economy-wide productivity growth under such circumstances, public expenditure as a share of GDP could decline significantly.

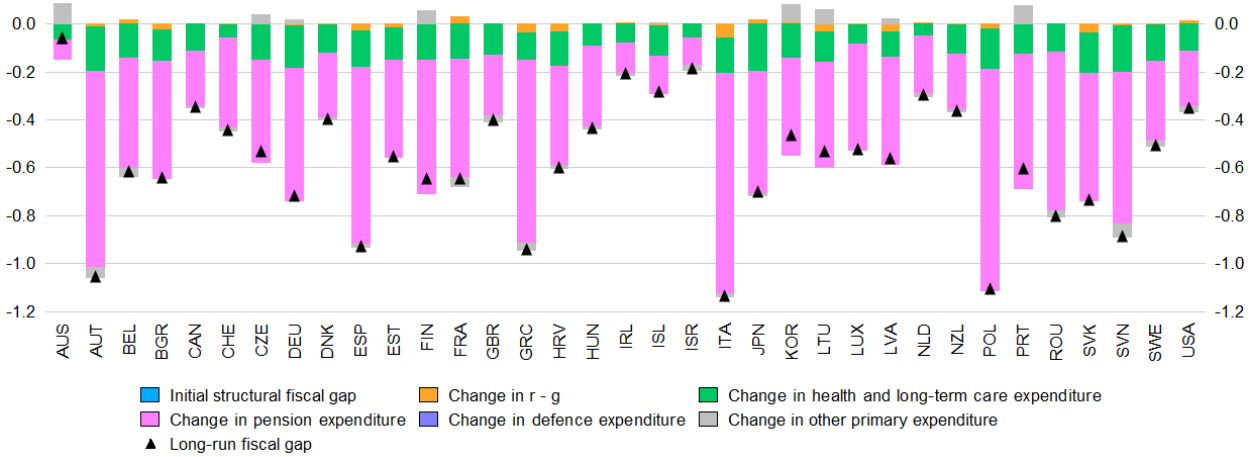
Sensitivity to net international migration

In the UNPD population projections used in the baseline scenario, country-level net international migration remains close to recent levels up to 2060. However, future population movements and migration policies are highly uncertain. To assess the sensitivity of long-run fiscal gap estimates to migration, an alternative scenario is run in which net migration in OECD countries rises over the baseline level by 0.1% of the current total population by 2033 and remains at that higher level thereafter, thus raising total population by about 3½ per cent over baseline in 2060 in the average country. In the LTM, new migrants are assumed to have the same economic characteristics (employment rates, average productivity level) as the native-born population, which could be an overoptimistic assumption in the short run even if justifiable in the longer term. Thus, migration affects long-run fiscal gaps solely through changes in the age structure of the population, with migrants assumed to be somewhat younger than the existing population. The calibration is based on the average age profile of immigrants in European countries. The scenario can therefore be interpreted as testing the sensitivity of long-run fiscal gaps to a younger age composition of the population.

Results show that increasing net migration by 0.1% of the population could reduce long-run fiscal gaps by 0.6 pp of GDP in the median country by 2060 (Figure 23). Most of this fiscal dividend comes through lower pension expenditure relative to GDP, as higher migration reduces the ratio of retirees to workers. It also lowers health and long-term care expenditure as a share of GDP (by reducing the proportion of the population aged 65 and over) and in many cases increases other current primary expenditure slightly (by increasing the ratio of youth to workers which is assumed to raise education spending as a share of GDP).

Figure 23. Impact on long-run fiscal gap of raising net international migration by 0.1% of population

Difference from baseline, % pts of potential GDP



Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

1. In practice, the fiscal effects of higher migration are more complex than can be illustrated with the LTM. Any downward impact on aggregate employment rates or average labour productivity would reduce the fiscal dividend associated with a younger population, as would the direct fiscal costs of administering a larger annual inflow of newcomers. At the same time, migration could raise employment rates and productivity, particularly if immigration policy targets high-skill migrants in occupations facing labour shortages and if newcomers receive adequate support to facilitate integration into the labour market, such as licensing assistance and language training.

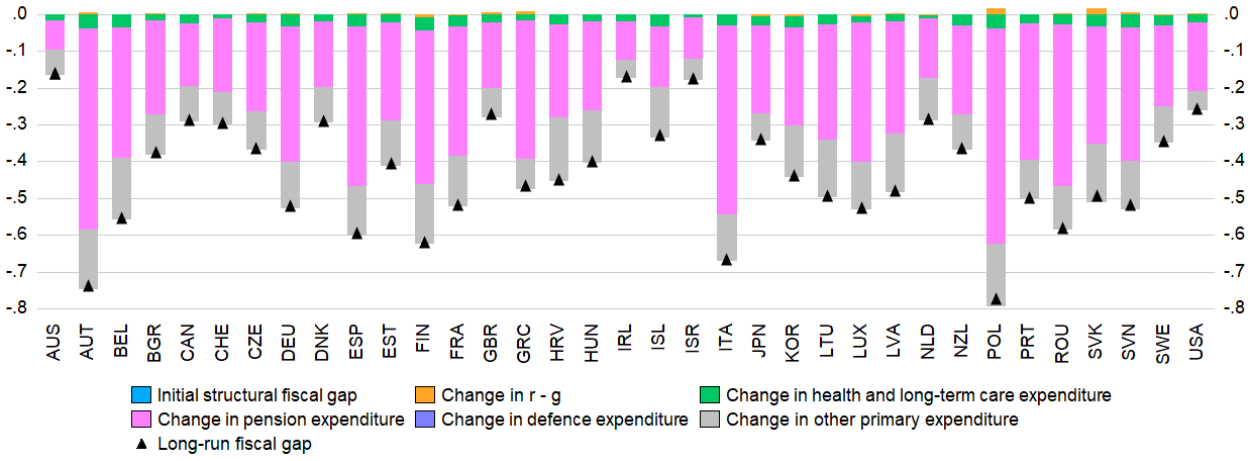
Sensitivity to pension age and indexation assumptions

The two key drivers of pension expenditure projections are the ratio of retirees to workers and the rate of benefit indexation. Two alternative scenarios are used to assess the sensitivity of future pension expenditure to these drivers.

The first assumes that all countries increase both male and female statutory retirement ages by one year by 2035, over and above any legislated increases already incorporated in the baseline. Such a reform reduces pension expenditure in 2060 by 0.3 pp of GDP in the median country (Figure 24). By raising employment rates among those aged 55-74 and thus increasing the employment-to-population ratio, the reform also lowers other current primary expenditure by a further 0.1 pp of GDP in the median country for the reasons explained in the higher-employment scenario (section 3.3). The total reduction in the long-run fiscal gap thus amounts to 0.4 pp of GDP in the median country, and more in countries where these expenditures are relatively high and/or where the retirees-to-workers and population-to-workers ratios are projected to increase the most (e.g. Austria, Poland).

Figure 24. Impact on long-run fiscal gap of raising statutory retirement ages by one year by 2035

Difference from baseline, % pts of potential GDP



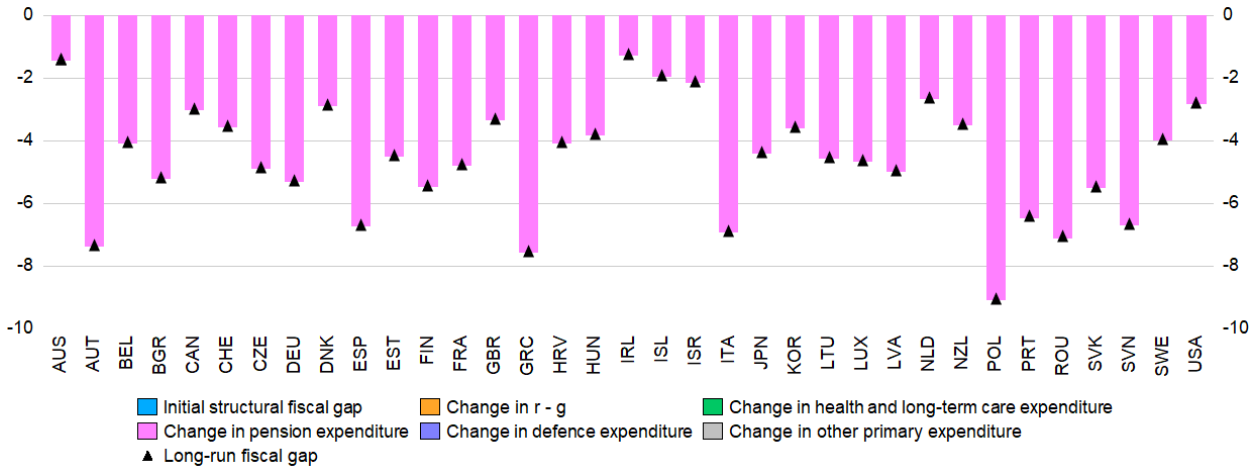
Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

The second scenario alters the benefit indexation assumption from wage to price indexation, resulting in a decline in the ratio of the average pension benefit to the average wage. This scenario has a very large effect on long-run fiscal gaps, reducing it by 4½ pp of GDP in the median country relative to baseline, and by as much as 7-9 pp in the most affected countries (Figure 25). As in the previous scenario, cross-country differences largely reflect baseline pension spending levels: countries with higher initial spending gain more (as a share of GDP) from reducing benefit generosity. A second factor is the projected average rate of labour productivity growth: the higher this rate, the larger the implied fall in the average pension benefit

ratio in this scenario, and therefore the larger the reduction in pension spending as a share of GDP. With price indexation, the ratio of pension expenditure to GDP would generally decline between 2027 and 2060 (by 1.7 pp of GDP in the median country) and the long-run fiscal gap of the median country would be only 0.6 pp of GDP.[12]

Figure 25. Impact on long-run fiscal gap of assuming price as opposed to wage indexation of pensions

Difference from baseline, % pts of potential GDP



Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

If pensions were indexed to prices as opposed to wages, it would be logical for this to also be the case for the rest of social protection expenditure (in the other primary expenditure component). Extending the price indexation assumption to the rest of social protection expenditure would generate substantial additional fiscal savings of 3 pp in the median country by 2060 relative to baseline.

While the potential fiscal savings from a reduction in social benefit generosity are substantial, it would undoubtedly be politically challenging. At the EU level for instance, it would imply that the average pension benefit ratio would fall by nearly one half, from 46% to 28% by 2060 given LTM projections for labour productivity growth. However, partial wage indexation, or temporary price indexation followed by a return to wage indexation, could still meaningfully reduce fiscal adjustment needs.

[12] The median long-run fiscal gap reported for this scenario (0.6 pp of GDP) does not correspond exactly to the difference between that of the baseline scenario (4.7 pp) and this section's price-indexation scenario (4½ pp) because medians are not additive.

4. Complementary illustrations of the need for long-run fiscal adjustments

This section illustrates long-run fiscal adjustment needs within the LTM using two approaches that complement the analysis in Section 2. Both confirm that, without policy action, many countries face sizeable risks to long-run fiscal sustainability.

Debt-to-GDP ratio trajectories

One way to highlight long-run fiscal challenges is to switch off the LTM's fiscal adjustment mechanism and project the debt-to-GDP trajectory that results from holding structural primary revenue as a percentage of potential GDP fixed at its initial level. Projected changes in spending, growth, and interest rates feed directly into the fiscal balance and the debt trajectory. Because the approach assumes no policy response to rising debt levels, some debt ratios begin to follow explosive paths. To avoid excessive feedback between debt and interest rates via fiscal risk premia, the illustrative simulation ends in 2045.

The projected change in the debt-to-GDP ratio is decomposed into components that are either identical or very similar to those used in section 2. The differences are as follows:

- 'Initial structural primary balance' replaces the previous 'initial structural fiscal gap' component. It shows the contribution to debt dynamics of maintaining the structural primary balance in 2027 unchanged as a percentage of GDP throughout the simulation period. It is simply the cumulative sum of this balance and is illustrative rather than aligned with official medium-term targets.
- The '$r - g$' component includes the effects of the initial level and changes in $r - g$ conditions interacting with debt and asset stocks. The feedback from rising debt to higher interest payments appears in this component.
- 'Other factors' consist mainly of the 'other primary expenditure' component, plus any cyclical effects that make the initial fiscal balance differ from the structural one in the first few years of the projection period.[13] These cyclical effects are generally very small.

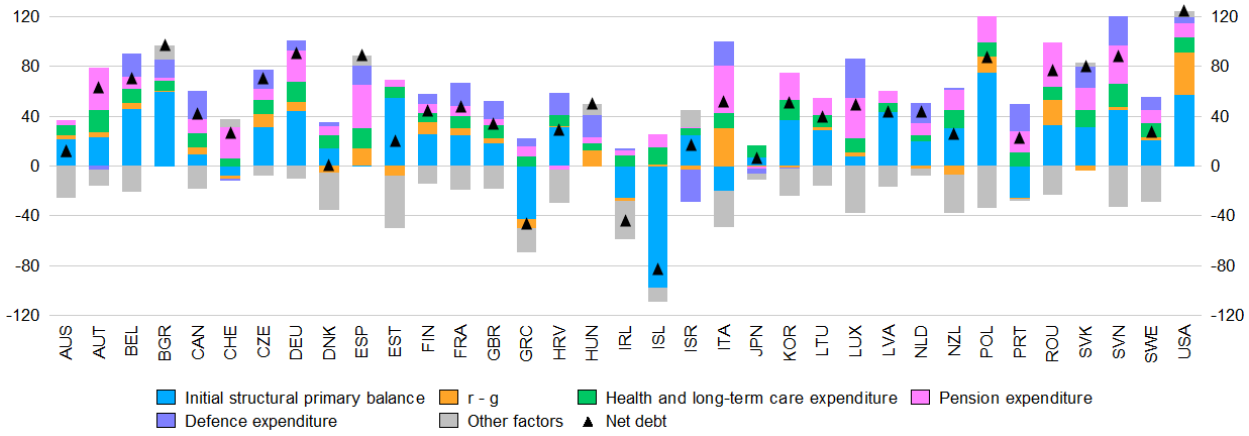
These illustrative projections show net government debt rising by 43 pp of GDP by 2045 in the median country, with the (assumed unchanged) initial structural primary balance being the largest contributor to the increase in debt (25 pp), followed by health and long-term care expenditure (10½ pp) and pensions (10 pp) (Figure 26). The projected increase in defence expenditure adds approximately 7¼ pp of GDP to net debt in the median country, again assuming no impact from defence spending on potential output. Other factors, mainly the reduction in primary expenditure other than for health, pension and defence explained in the previous section, subtract about 16 pp of GDP in debt in the median country.[14]

[13] Unlike the analysis in section 2, which decomposes a cyclically-adjusted variable, the debt-to-GDP ratio is affected by cyclical effects on GDP, making this addition to the decomposition necessary.

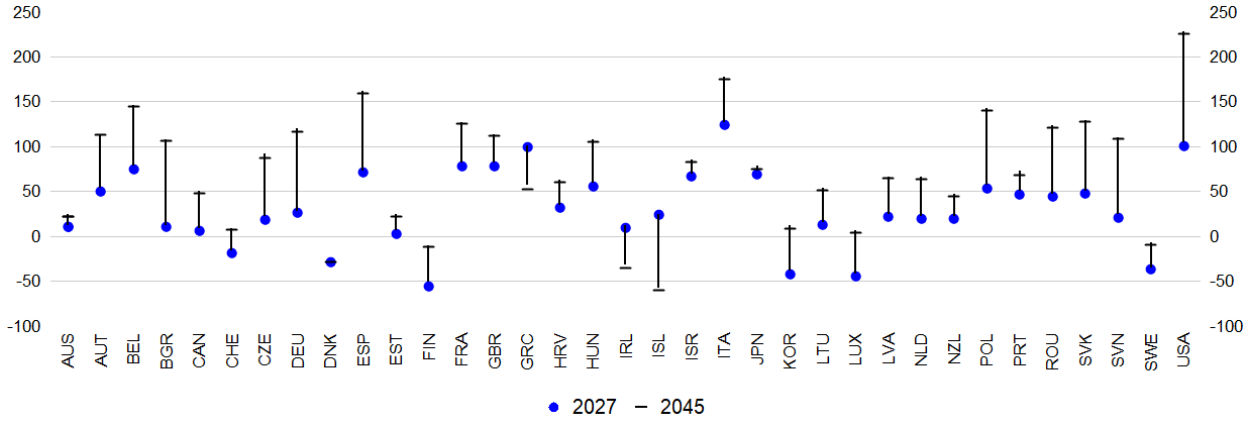
[14] The sum of median components does not add up to the reported total exactly because medians are not additive.

Figure 26. Net government debt with fiscal adjustment rule switched off, 2027-45

A. Change between 2027 and 2045, % pts of GDP



B. Levels, 2027 and 2045, % of GDP



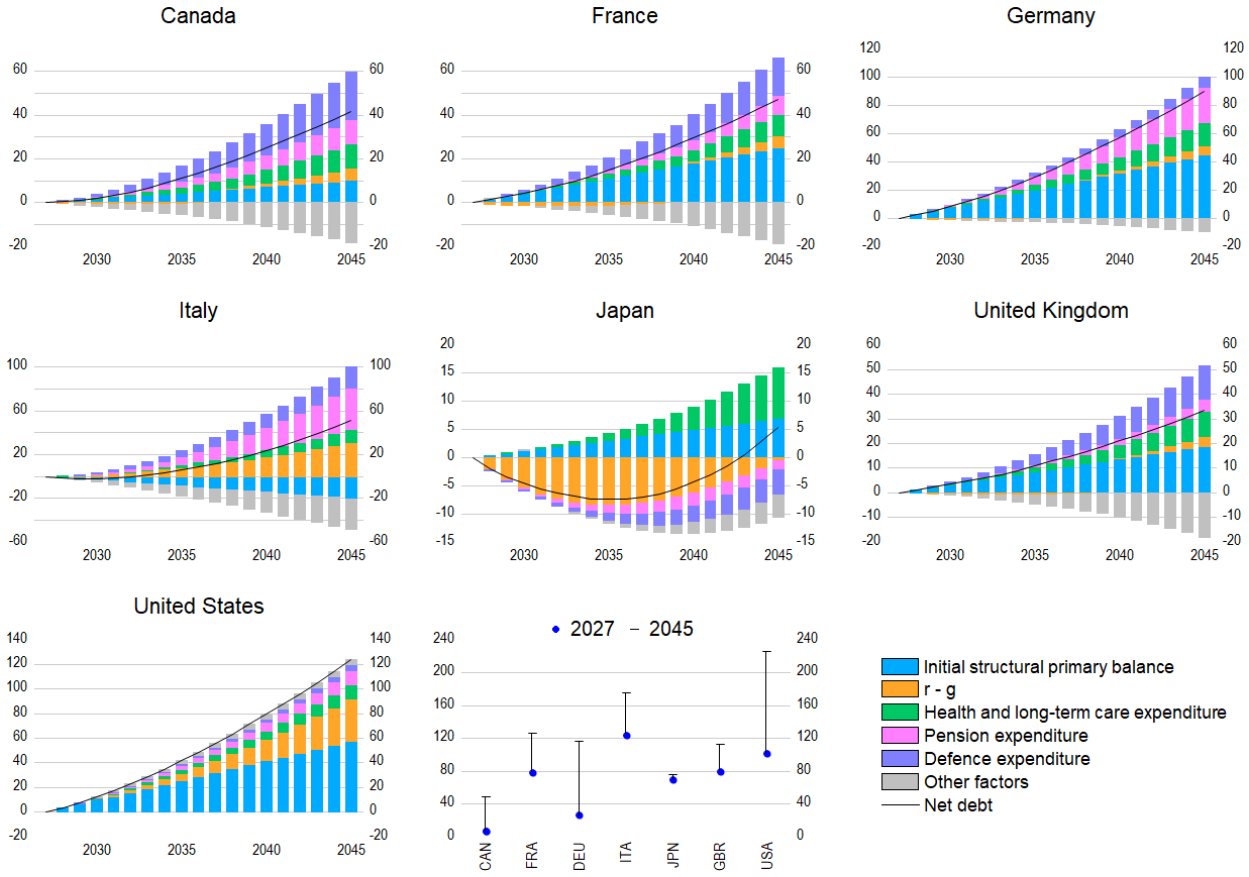
Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

Across G7 countries, government net debt increases by between 5½ pp of GDP in Japan to 124 pp of GDP in the United States (Figure 27; projections for other countries appear in Annex C). A few results stand out:

- If left unchanged, the negative initial structural primary balances in France, the United Kingdom, and the United States could add more to debt than the projected increases in health, long-term care, and pension spending.
- Similarly, in Canada, France, and the United Kingdom, meeting NATO defence spending commitments could add roughly as much, or more, to debt as rising spending on health, long-term care, and pensions.

Figure 27. Change in net government debt from 2027 with the fiscal adjustment rule switched off

% pts of GDP



Note: The projections use a common framework for all countries, including an assumption that government financial assets remain constant as a share of GDP in all countries. Such assumptions may depart substantially from national objectives and projections in particular economies. For instance, in Canada, official projections anticipate significant growth in pension plan (CPP/QPP) assets relative to GDP. If fully reflected on a consolidated general government basis, this accumulation would result in a lower net debt-to-GDP trajectory than shown in this chart. Such issues are best addressed on a country-by-country basis, particularly in the context of OECD country surveys.
Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

The projected decline in Japan's debt ratio early in the period is noteworthy. It reflects its initially favourable $r^d - g$ differential combined with a very high gross debt ratio, which magnifies this effect. However, as the implicit average interest rate gradually converges towards an equilibrium somewhat above the potential growth rate (see section 2.4), this trend reverses midway through the projection period. By 2060, the cumulative contribution of the $r - g$ component would again be positive — consistent with the longer-horizon results shown by the standard long-run fiscal gap indicator (Figure 2, Panel A).

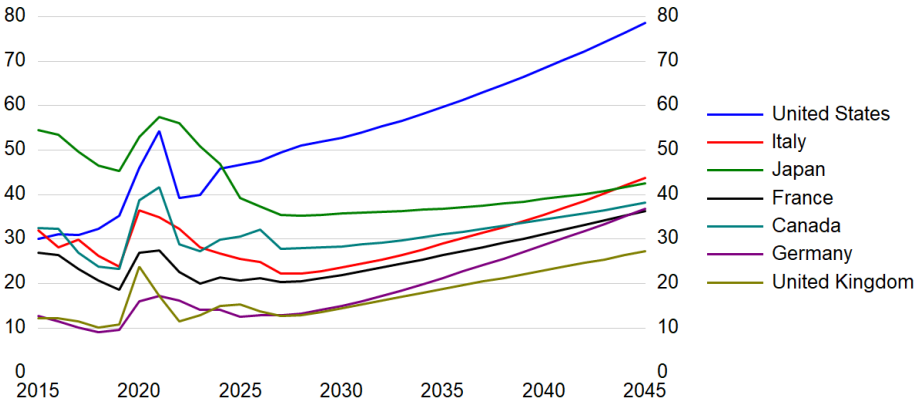
Gross government financing needs

Another useful indicator of fiscal challenges is the projected evolution of general government gross financing needs, defined as the sum of new borrowing and maturing debt. After spiking during the pandemic, annual financing needs now stand at around 25% of GDP in the median G7 country. In the scenario described in the previous subsection where budget balances are allowed to deteriorate in line with spending pressures, financing needs rise sharply in the coming years, particularly in the United States (due to a large initial primary deficit) and Italy (due to high initial debt) (Figure 28, projections for other countries appear in Annex C). The United Kingdom is partially insulated by its relatively long debt maturity

profile (an average remaining maturity of around thirteen years, compared with about seven years for the rest of the G7 as of Q1 2026). The projections and maturity statistics do not account for substantial central bank holdings of government debt, which effectively shorten the consolidated public sector's average maturity, by around two years in the average economy (OECD, 2023[24]). Future changes in central bank balance sheet policy could affect the market interest rate on new debt, but are not included in this exercise.

Figure 28. Projected annual gross government financing needs

% of GDP



Source: OECD Economic Outlook 119 long-term scenarios database and author's calculations.

5. The energy transition is potentially an additional source of expenditure pressure

The energy transition, broader climate change mitigation efforts, and fiscal costs related to climate adaptation and climate-related damages are additional sources of potential fiscal pressure that are only partially included in the LTM and not included in the long-run fiscal gap indicator. This section updates and refines the estimates of projected capital expenditure for renewable electricity generation first presented in Guillemette and Château (2023[2]).

The LTM's baseline scenario is built on a business-as-usual premise, including for the relationship between economic output and energy use and for the gradual transition from carbon-based to low-carbon energy sources. The energy-substitution assumption is implemented by extrapolating country-level primary energy mixes along time trends estimated over the past decade. This approach results in a projected pace of substitution that is insufficient to achieve global net-zero greenhouse gas (GHG) emissions this century: carbon-based sources still account for roughly three-quarters of the global primary energy mix in 2050 and more than 60% in 2100.

The LTM can also be used to generate other stylised energy-transition scenarios with much more ambitious substitution and efficiency gains. In the scenario used below, all countries accelerate decarbonisation, phasing out coal by 2050 at the latest and reducing the shares of oil and gas in primary energy to 5% and 10%, respectively, by then. These declining shares are replaced by low-carbon sources, primarily solar and wind, with the allocation varying by country according to stylised projection rules that reflect physical constraints and revealed preferences, albeit imperfectly.[15] This scenario limits global warming to about

[15] See section 2.1 of Annex B in Guillemette and Château (2023[2]) for how this is done. While many different configurations of individual country energy mixes, energy use, etc., could be consistent with the Paris agreement goals,

1.7°C by 2100, consistent with the Paris Agreement. With a few exceptions (e.g. geothermal heat), low-carbon energy sources generate electricity. The transition therefore requires a substantial expansion of electricity-generation capacity. Using the LTM's projections for total primary energy use and energy mixes, indicative projections can be derived for the additional electrical capacity required in both the baseline and transition scenarios.[16]

These projections are highly uncertain and cover only a narrow aspect of the fiscal implications of climate change and the energy transition.[17] They are intended to provide broad quantification and cross-country comparisons on this one aspect, namely the capital costs associated with transitioning to low-carbon energy, not to give a full picture of the potential fiscal challenges associated with the climate/energy nexus. This focus allows the calculations to ignore the choice of policy instruments to drive the transition (e.g. a carbon tax), any associated revenue, and the possibility of a near-term negative supply shock that could depress output relative to baseline. The projections also omit many country specificities. For example, capacity-factor assumptions for different energy technologies are held constant across countries (Table 4), whereas in reality they vary with geography and climate.

Table 4. Assumptions for new low-carbon electricity generation capacity

	Average capacity factor	Average building cost of new capacity in 2024, USD billion per GW				Assumed learning rate, % decline in real building cost with each doubling of installed capacity
		EU	USA	Japan	Middle East	
Wind	0.35	1.8	1.8	3.2	1.7	6
Solar	0.17	0.8	1.3	2.0	0.7	20
Hydro	0.48	3.3	3.4	3.0	2.8	1
Nuclear	0.85	6.6	5.0	4.2	3.5	0
Other renewables	0.78	3.7	3.7	3.7	3.3	5

Source: Capacity factors are based on global 2024 averages for new capacity reported by IRENA (2025), *Renewable power generation costs in 2024*, International Renewable Energy Agency, Abu Dhabi. Cost estimates and learning rates are based on IEA (2025), *World Energy Outlook 2025*, International Energy Agency, Paris. The estimates for wind include onshore/offshore wind in a 90/10 proportion. The estimates for solar includes solar photovoltaic and concentrated solar power in a 99/1 proportion. The estimates for other renewables are simple averages of the six different sub-types (bioenergy, geothermal power, etc.). In the absence of specific estimates for these countries from the IEA, the EU cost estimates are used for Australia and New Zealand, and the Japan cost estimates are used for Korea.

In the energy-transition scenario, installed low-carbon generation capacity is projected to increase by a factor of four to ten by 2050 in most countries (Figure 29). The increase is smaller in countries where low-carbon sources already account for a large share of primary energy (e.g. Iceland) and larger in countries with high carbon intensity. Solar dominates new capacity additions, partly by assumption, given its rapidly declining costs, and partly because its relatively low capacity factor implies a larger installed base is

the specific assumptions implemented here are guided by simplicity and similarity of treatment, so that results are comparable across countries and hopefully illustrative in terms of broad orders of magnitude. They do not necessarily adhere to individual country commitments, targets or constraints, but this can be refined when the model is applied to individual country analyses.

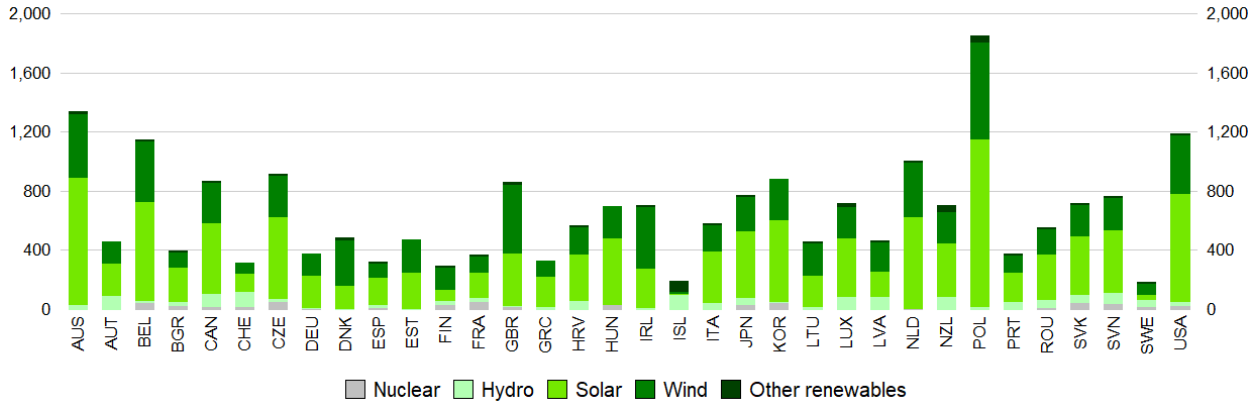
[16] See section 2.5 of Annex B in Guillemette and Château (2023[2]) for details on how this is done.

[17] See https://www.oecd.org/en/data/tools/oecd-tool-for-calculating-the-fiscal-impacts-of-decarbonisation-and-environmental-challenges.html for a tool the OECD has developed to evaluate the potential fiscal consequences of environmental pressures and decarbonization more broadly, and in particular the potential loss of fuel excise tax revenue associated with electrification of the vehicle fleet.

required for any given level of annual generation. While highly stylised, these projections illustrate the scale of the challenge in expanding low-carbon electricity.

Figure 29. Increase in low-carbon electrical capacity by 2050 in the energy transition scenario

Index of low-carbon electricity-generation capacity, 2023 = 100



Note: Israel is not shown as it is an outlier given very low capacity in the base year (2023).
Source: IEA Electricity Information Statistics database (for the base year) and OECD Economic Outlook 119 long-term scenarios database.

Combining the capacity projections with cost estimates by technology allows indicative projections of future capital investment needs. For most countries initial cost assumptions are based on regional averages and do not capture country-specific conditions perfectly (Table 4). Future cost declines are also uncertain: they are assumed to follow a power-law learning curve in which real plant costs (relative to the GDP deflator) fall by some percentage with each doubling of installed global capacity (Table 4).[18] Linking country-specific costs to global installed capacity generates a positive externality: faster global deployment accelerates cost reductions. Since low-carbon capacity expands more rapidly in the transition scenario than in baseline, real costs fall more quickly.

Capital investment in generation capacity is only one component of total investment needs. A renewable-heavy grid also requires major investment in transmission lines, transformers, and other equipment, as well as new technologies, such as storage, to manage intermittency. Based on European averages, for every dollar invested in renewable generation, around 70 cents is required for grid upgrades (BloombergNEF, 2025[25]). Accordingly, the LTM's estimated capital costs for new low-carbon generation are multiplied by 1.7 to approximate total investment needs for low-carbon electricity systems. This still excludes many other necessary investments and does not capture differences in operating costs across technologies.

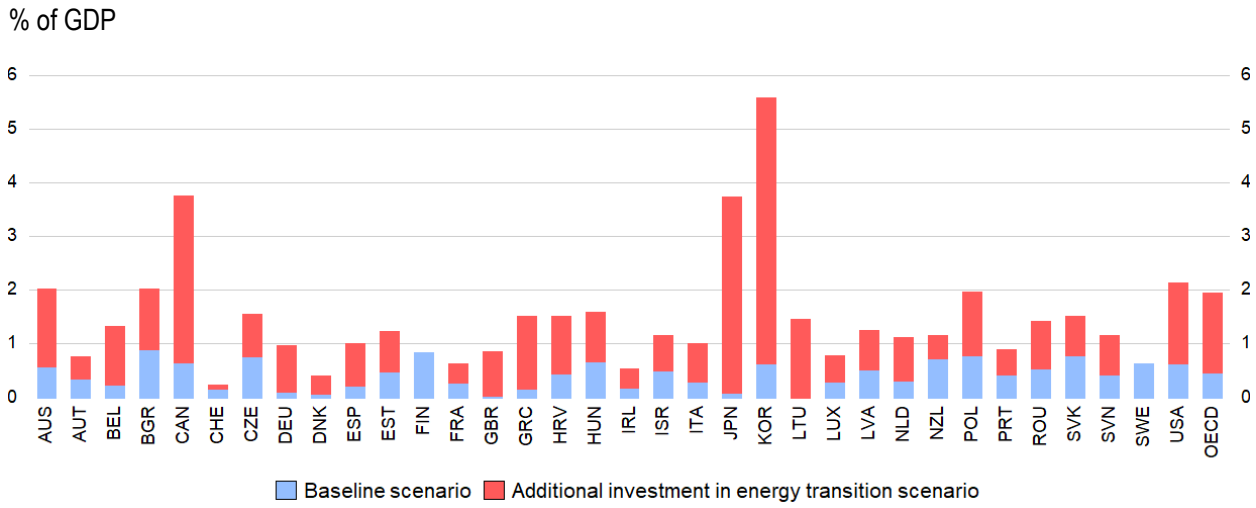
Investment needs under the transition scenario cannot simply be added to the long-run fiscal gap indicator for several reasons. First, low-carbon electricity investment does not necessarily have to be financed by governments, even though electricity supply has historically involved substantial public financing. Second, if the transition is completed by 2050, investment thereafter would fall back toward baseline levels, making

[18] The literature provides various estimates for learning rates in renewable power, with some studies pointing to even higher learning rates than these assumptions over the last few years (IRENA, 2025[30]). However, learning rate estimates often refer to equipment prices (i.e., the solar panels themselves), not necessarily the full installed system cost (which includes labour for installation, etc.). Learning rates for overall system costs are likely to be lower than purely for equipment. Because data for many non-OECD countries are limited, total installed capacity in the OECD area is used in the LTM as a proportionate proxy for global capacity.

the 2060 horizon used elsewhere in this paper not well adapted to this specific spending pressure. Moreover, achieving the Paris objective requires front loading investment, given long planning and construction periods. Early-period investment is also less uncertain than projections several decades out. For these reasons, investment needs are reported for 2028-2035 (the first eight years of the long-run simulation) and are not added to the long-run fiscal gap metric for 2060 elsewhere in the paper.

In the energy-transition scenario, new investment in low-carbon electricity capacity amounts to 2% of GDP in the OECD area between 2028 and 2035 (Figure 30). Differences in initial energy mixes and in cost estimates for new capacity largely explain cross-country variation — for example, euro area countries generally show lower investment needs than other OECD economies both because of greener initial energy mixes and lower costs for new solar installations. Because the baseline also includes some investment in new capacity, these costs do not represent a pure increase: the transition scenario adds 1½ pp of GDP to baseline investment needs for the OECD area, and as much as 3½-5 pp in Japan and Korea given relatively high costs for new capacity in this region. If governments finance a large share of this investment, the transition could add several tenths of a percentage point to expenditure pressures in the early projection years, and perhaps a few percentage points of GDP in some countries.

Figure 30. Projected capital expenditure for low-carbon electrical capacity and grid updating, 2028-to-2035 average



Note: Iceland is not shown as its very peculiar energy mix, with a lot of geothermal energy already, implies no need for additional investment in the transition scenario.
Source: OECD Economic Outlook 119 long-term scenarios database.

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Annex A. Decomposition of the long-run fiscal gap indicator

This Annex shows the algebraic decomposition of the long-run fiscal gap shown in Figure 2. The variable names follow *OECD Economic Outlook* (as well as LTM) mnemonics wherever possible (Table A A.1).

Table A A.1. Variable names and definitions

Economic Outlook Long-Term database mnemonic	Definition	Shorthand used below if applicable
CAPOGQ	Net capital outlays of the government, % of potential GDP	
CAPOGQ_DEF	Net capital outlays of the government on defence, % of potential GDP	
CAPOGQ_EXDEF	Net capital outlays of the government excluding those on defence, % of potential GDP	
DEFENCE	Defence expenditure, % of potential GDP	
GDPTR	Potential gross domestic product, nominal, local currency	$g = \Delta\log(GDPTR_t)$
GFAR	Government gross financial assets, % of GDP	
GGFLQ	Government gross financial liabilities, % of GDP	
HEALTH	Public expenditure on health and long-term care, % of potential GDP	
NLGXQA	Cyclically adjusted government primary balance, % of potential GDP	
PENSION	Public pension spending, % of potential GDP	
RATE_GGINTP	Implicit interest rate paid on gross government financial liabilities, %	$r^d = \frac{RATE_GGINTP}{100}$
RATE_GGINTR	Implicit interest rate received on government financial assets, %	$r^a = \frac{RATE_GGINTR}{100}$
YPGXQA	Cyclically adjusted current government disbursements excluding interest, % of potential GDP	
YPGXXQA	Cyclically adjusted current government disbursements excluding interest, health and long-term care, pensions and defence, % of potential GDP	
YRGXQA	Cyclically adjusted current government receipts excluding interest, % of potential GDP	

As explained in the main text, the long-run fiscal gap is defined as the change in cyclically-adjusted primary government revenue in year t ($YRGXQA_t$) over a chosen period, 2027 to 2060 being used in the main text. Using the notation $\Theta X = X_{2060} - X_{2027}$, this is:

$$\Theta YRGXQA = YRGXQA_{2060} - YRGXQA_{2027} \quad [A1]$$

To stabilise the debt-to-GDP ratio at target over the projection period, the LTM dynamically adjusts its fiscal instrument, $YRGXQA_t$, so that it matches a target value, $YRGXQA_t^*$, that is consistent with maintaining the debt-to-GDP ratio stable given other variables in the model.

This target value is derived from the conventional debt dynamics equation but using trend values (such as potential output and cyclically-adjusted fiscal aggregates) given that, in the LTM, the output gap is assumed to close within a few years of the start of the projection period, from which point on fiscal aggregates are

equal to their cyclically-adjusted values. The equation also accounts for government financial assets ($GFAR$):

$$\Delta GGFLQ_t = \left(\frac{r_t^d - g_t}{1+g_t}\right) \cdot GGFLQ_{t-1} + \left(\frac{g_t/2 - r_t^a}{1+g_t}\right) \cdot GFAR - NLGXQA_t \quad \text{[A2]}$$

$GFAR$ has no time subscript because it is assumed to remain constant over the projection period. The g_t in the numerator of the second term is divided by two because of the EO/LTM assumption, explained in the main text, that half of the annual increase in government financial assets (to keep pace with GDP) comes from revaluation effects and therefore requires no debt issuance.

Setting the left-hand side of [A2] to zero and moving the rightmost term to that side of the equation, a target cyclically-adjusted primary balance ($NLGXQA_t^*$) can be defined to achieve the objective of maintaining the gross debt ratio stable as a function of the target level ($GGFLQ^*$):

$$NLGXQA_t^* = \left(\frac{r_t^d - g_t}{1+g_t}\right) \cdot GGFLQ^* + \left(\frac{g_t/2 - r_t^a}{1+g_t}\right) \cdot GFAR \quad \text{[A3]}$$

This equation is the basis for the first two components of the long-run fiscal gap decomposition. The 'initial structural fiscal gap' ($ISFG_{2027}$) is defined as the difference between the target value given by [A3] and the actual value of this variable in the initial year:

$$ISFG_{2027} = NLGXQA_{2027}^* - NLGXQA_{2027} \quad \text{[A4]}$$

The 'change in r – g' component is defined as the change in [A3] over the period considered, or $\Theta NLGXQA^*$. Since the stock variables in [A3] are constants, this quantity is only a function of projected changes in r_t^d, r_t^a and g_t, as intended.

Coming back to the target value $YRGXQA_t^*$, it is defined using the accounting identity for $NLGXQA_t$ re-arranged to put $YRGXQA_t$ on the left-hand side and replacing $NLGXQA_t$ with the just-defined target value:

$$YRGXQA_t^* = NLGXQA_t^* + YPGXQA_t + CAPOGQ_t \quad \text{[A5]}$$

By identity, then, when $YRGXQA_t$ is at this target value, $NLGXQA_t$ is also at its target value. The fiscal reaction function in the LTM is designed so that $YRGXQA_t$ goes toward $YRGXQA_t^*$. This is not always exactly true given that a limit is imposed on the annual change in $YRGXQA_t$ to reflect real-world political-economy constraints on the size of any annual fiscal adjustment and also because $YRGXQA_t^*$ fluctuates from year to year. But in principle, it is very close to its target value in the long run, thus $YRGXQA_{2060} = YRGXQA_{2060}^*$, and [A1] can be re-written:

$$\Theta YRGXQA = YRGXQA_{2060}^* - YRGXQA_{2027} \quad \text{[A6]}$$

Using the same identity that is the basis for [A5] to expand the two terms, this becomes:

$$\begin{aligned} \Theta YRGXQA &= NLGXQA_{2060}^* + YPGXQA_{2060} + CAPOGQ_{2060} - (NLGXQA_{2027} + YPGXQA_{2027} \\ &\quad + CAPOGQ_{2027}) \\ &= NLGXQA_{2060}^* - NLGXQA_{2027} + \Theta YPGXQA + \Theta CAPOGQ \end{aligned} \quad \text{[A7]}$$

Using [A4] to substitute for $NLGXQA_{2027}$, this becomes:

$$\begin{aligned} \Theta YRGXQA &= NLGXQA_{2060}^* - (NLGXQA_{2027}^* - ISFG_{2027}) + \Theta YPGXQA + \Theta CAPOGQ \\ &= ISFG_{2027} + \Theta NLGXQA^* + \Theta YPGXQA + \Theta CAPOGQ \end{aligned} \quad \text{[A8]}$$

Government net capital expenditure is composed of defence and a residual (ex-defence) component:

$$CAPOGQ_t \equiv CAPOGQ_DEF_t + CAPOGQ_EXDEF_t \quad [A9]$$

The variable $YPGXXQA_t$ is not a conventional national accounts concept. It is defined for the purposes of the LTM as the remaining current primary expenditure after deducting those components that are projected explicitly:

$$YPGXXQA_t \equiv YPGXQA_t - PENSION_t - HEALTH_t - (DEFENCE_t - CAPOGQ_DEF_t) \quad [A10]$$

with the term in parenthesis measuring current defence expenditure. Using [A9] and [A10] (re-arranged) to expand $\theta YPGXQA$ in [A8]:

$$\theta YRGXQA = ISFG_{2027} + \theta NLGXQA^* + \theta PENSION + \theta HEALTH + \theta DEFENCE + \theta YPGXXQA + \theta CAPOGQ_EXDEF \quad [A11]$$

Equation [A11] gives the exact decomposition of the long-run fiscal gap (Table A A.2). Any discrepancy between the right-hand and left-hand sides is attributable to the condition $YRGXQA_{2060} = YRGXQA^*_{2060}$ not holding exactly as the mechanics of the fiscal reaction function in the LTM can make the instrument depart from the moving target slightly in any given year for the reasons explained previously.

Table A A.2. Decomposition of the long-run fiscal gap

Component as shown in Figure 2, Panel A	Term in equation [A11]
Initial structural fiscal gap	$ISFG_{2027}$
Change in *r* - *g*	$\theta NLGXQA^*$
Change in pension expenditure	$\theta PENSION$
Change in health and long-term care expenditure	$\theta HEALTH$
Change in defence expenditure	$\theta DEFENCE$
Change in other primary expenditure	$\theta YPGXXQA + \theta CAPOGQ_EXDEF$
Long-run fiscal gap	$\theta YRGXQA$

Annex B. Projection of public pension expenditure

The approach to projecting public pension expenditure as a percentage of potential GDP ($PENSION_t$) has not fundamentally changed from the approach described in Guillemette (2019[26]). It starts from the following identity, with time subscript omitted:

$$PENSION \equiv \frac{PPE}{GDPTR} \equiv \frac{PENSIONERS}{ETPT} \cdot \frac{AVG_PENSION}{AVG_WAGE} \cdot \frac{LAB_INCOME}{GDPTR} \quad [B1]$$

where PPE is total of old-age and survivor benefits in local currency, $GDPTR$ is nominal potential gross domestic product also in local currency, $PENSIONERS$ is the number of people receiving a pension, $ETPT$ is potential employment, $AVG_PENSION$ is the average pension benefit, AVG_WAGE is the average wage among workers and LAB_INCOME is potential total labour income. The identity holds because $LAB_INCOME = ETPT \cdot AVG_WAGE$ while $PPE = PENSIONERS \cdot AVG_PENSION$. Taking logs and time differencing [B1] yields an expression for the growth rate of public pension expenditure as a share of potential GDP:

$$\Delta log(PENSION) \equiv \Delta log\left(\frac{PENSIONERS}{ETPT}\right) + \Delta log\left(\frac{AVG_PENSION}{AVG_WAGE}\right) + \Delta log\left(\frac{LAB_INCOME}{GDPTR}\right) \quad [B2]$$

The first term of this expression is the rate of change in the ratio of pensioners to workers. This ratio is not available per se in the LTM, but for projection purposes its evolution is approximated by the following expression:

$$\Delta log\left(\frac{PENSIONERS}{ETPT}\right) \cong \Delta log\left(\frac{(1 - ER_{5574}) \cdot POP_{5574} + POP_{75+}}{ETPT}\right) \quad [B3]$$

where $ER_{5574}$is the employment rate for 55-to-74-year-olds (in decimals), POP_{5574} is the population in this age range and POP_{75+} is the population aged 75 and up. This expression appears to imply that non-employed people between the ages of 55 and 74 and all those aged 75 and above are pensioners, which is never strictly the case. But it is only the log *change* in the expression that matters to the projection. The idea is that the change in this ratio is a good proxy for the proportionate change in the true ratio of pensioners to workers.

The second term of [B2] is the rate of change in the ratio of the average pension to the average wage, also called the average benefit ratio (not to be confused with the replacement rate, which usually refers to a new retiree's initial pension as a percentage of her last salary). Its numerator, the average pension, aggregates over all retirees and all types of pensions at a given time. As such, its future evolution depends on many factors, including eligibility rules for new pension benefits, the method used to uprate past earnings in the benefit formulae of defined-benefit plans, the rate of indexation of benefits, the age structure of the population of benefit recipients, rates of return on plan assets, etc. Projecting the future evolution of benefit ratios thus requires a large amount of information, not just on current rules, but also on how they will evolve given the many reforms that have already been legislated in many countries.

Under the simplifying assumption that the indexation rule for existing benefits alone drives the evolution of the benefit ratio over time, this evolution can be connected to the projected trend labour productivity growth rate. In countries where public pension benefits are indexed to the consumer price index, the average benefit ratio would decline each year by the wedge between nominal wage growth and consumer price inflation (i.e. real wage growth). In the LTM, real wage growth is equal to trend labour productivity growth by construction, so with price indexation, the average benefit ratio falls at the rate of trend labour productivity growth. In countries where public pension benefits are indexed to wages, the average benefit ratio would remain constant, so the second term of [B2] would always be zero. Some countries use a formula, or a series of discrete adjustments, to maintain the indexation of public pension benefits somewhere between price indexation and full wage indexation.

For the LTM to be able to reflect a range of assumptions, the following projection rule is used:

$$\Delta log\left(\frac{AVG_PENSION_t}{AVG_WAGE_t}\right) = (\varphi - 1) \cdot \Delta\log(TRPDTY_t) \quad \text{[B4]}$$

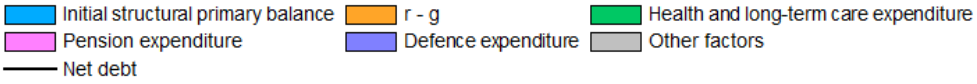
where φ is a coefficient reflecting the degree to which public pensions are indexed to wages and $TRPDTY_t$ is trend labour productivity. As explained in the main text, the LTM's business-as-usual scenario assumes a stable average benefit ratio, both for simplicity and based on the historical experience of EU countries. This is equivalent to assuming full wage indexation of pension benefits, or $\varphi = 1$. Setting $\varphi = 0$ would imply price indexation. Partial wage indexation can be simulated with φ somewhere between zero and one.

Finally, the third term of [B2] is the rate of change in the labour share of income. In the LTM, the labour share of income is assumed constant, so this term is zero. In view of the general decline of labour income shares in recent years (Schwellnus et al., 2018[27]), as well as evidence of policy influences behind this trend (Pak and Schwellnus, 2019[28]), the assumption of a fixed labour share of income may be revisited in future model development.

Annex C. Individual country charts for net debt and gross financing needs projections

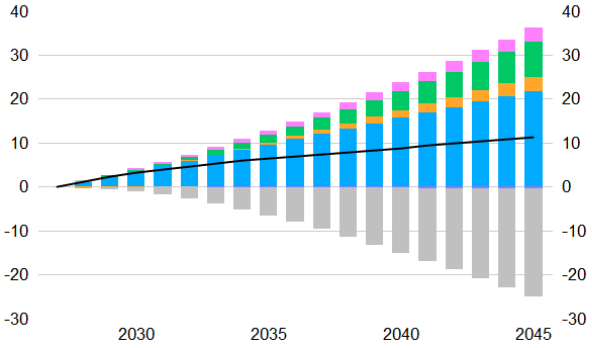
This Annex complements sections 4.1 and 4.2 of the main text, showing the projected change from 2027 in net government debt and gross financing needs for non-G7 countries in the scenario where the LTM's fiscal adjustment rule is switched off.

Legend for decomposition of change in net government debt:

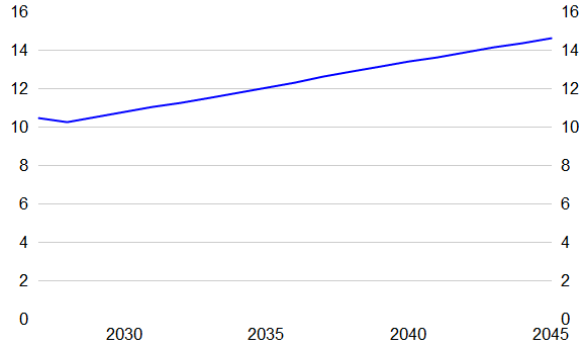


AUS - Australia

Change in net debt from 2027, % pts of GDP

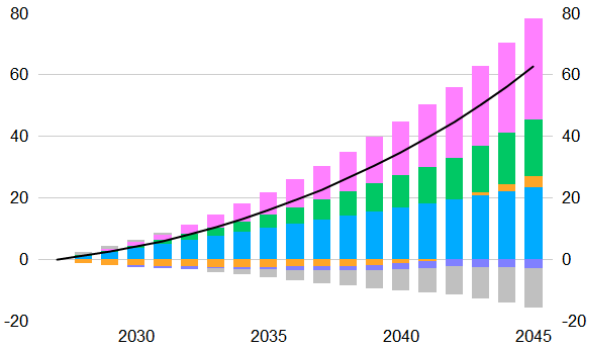


Gross financing needs, % of GDP

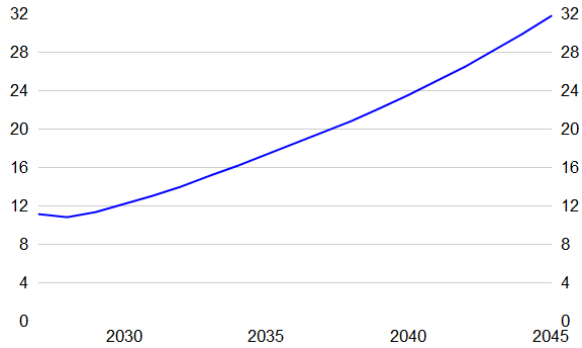


AUT - Austria

Change in net debt from 2027, % pts of GDP

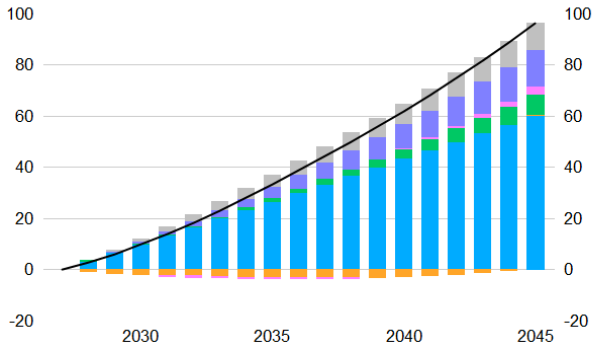


Gross financing needs, % of GDP

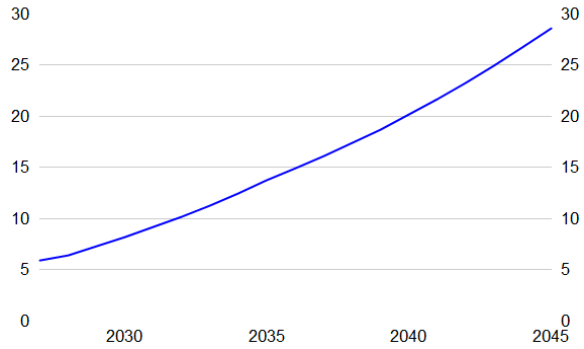


BGR - Bulgaria

Change in net debt from 2027, % pts of GDP

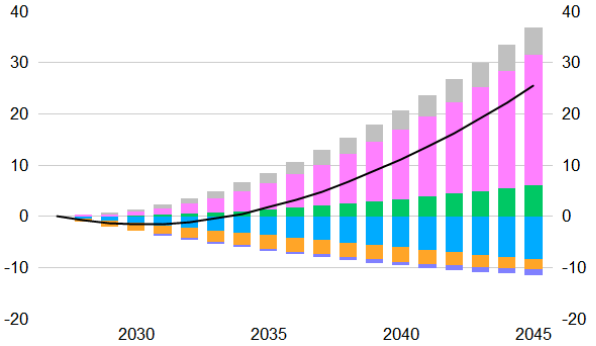


Gross financing needs, % of GDP

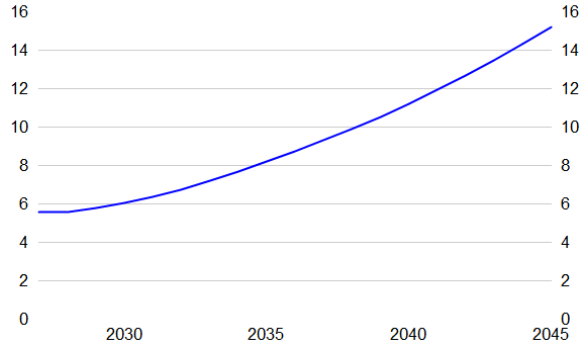


CHE - Switzerland

Change in net debt from 2027, % pts of GDP

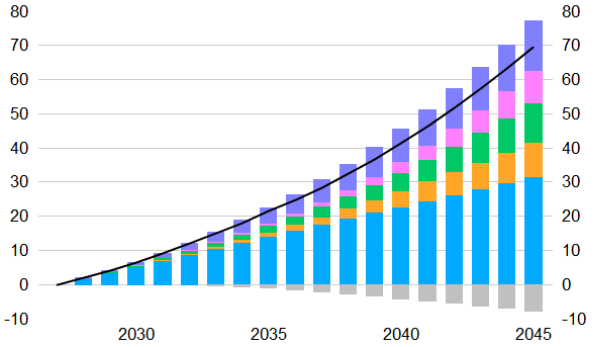


Gross financing needs, % of GDP

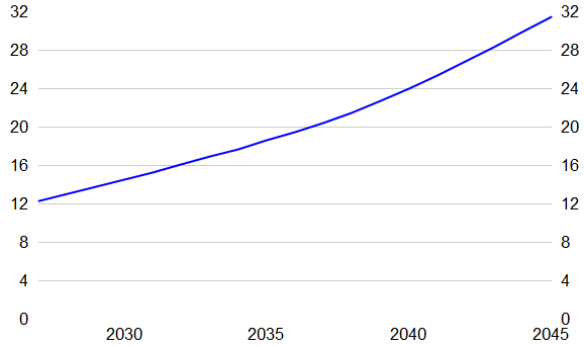


CZE - Czechia

Change in net debt from 2027, % pts of GDP

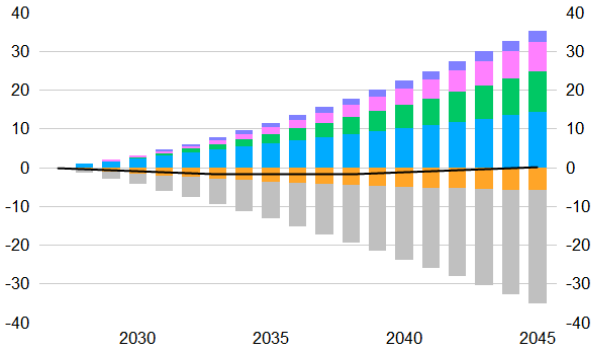


Gross financing needs, % of GDP

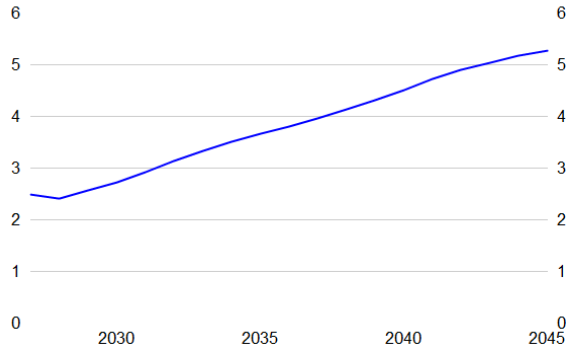


DNK - Denmark

Change in net debt from 2027, % pts of GDP

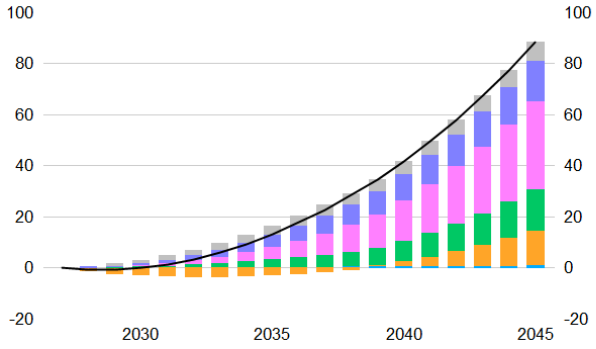


Gross financing needs, % of GDP

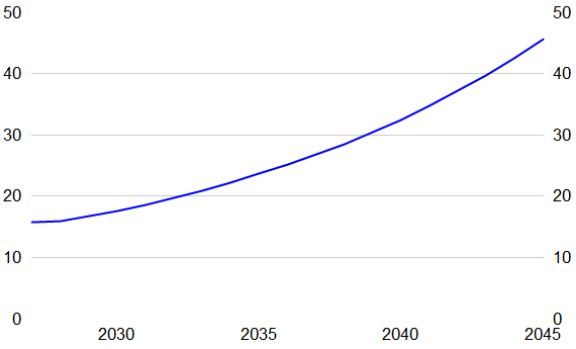


ESP - Spain

Change in net debt from 2027, % pts of GDP

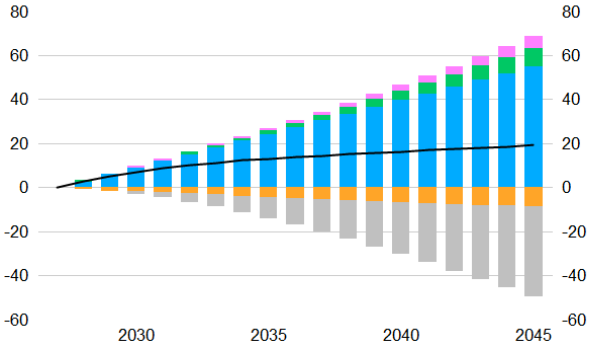


Gross financing needs, % of GDP

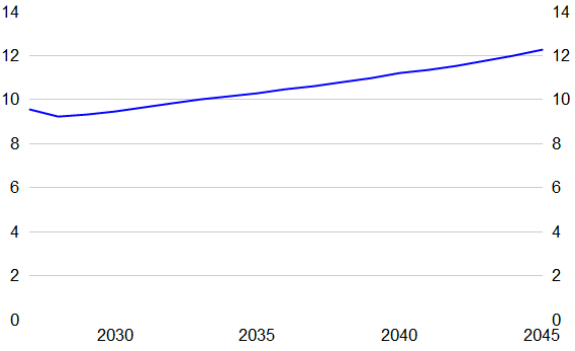


EST - Estonia

Change in net debt from 2027, % pts of GDP

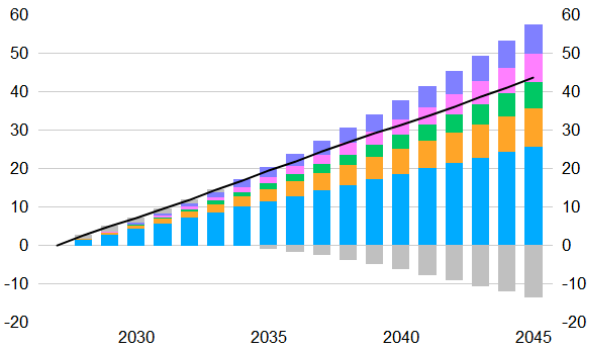


Gross financing needs, % of GDP

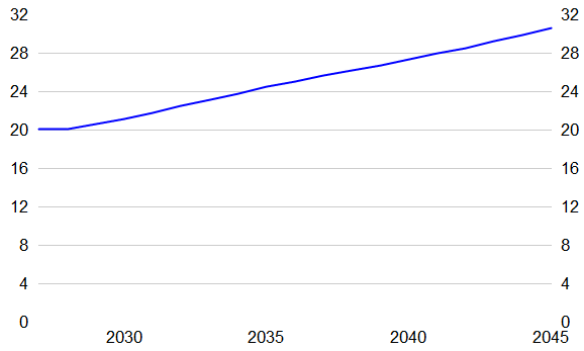


FIN - Finland

Change in net debt from 2027, % pts of GDP

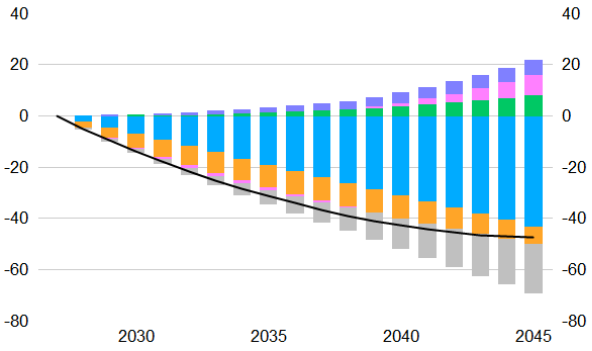


Gross financing needs, % of GDP

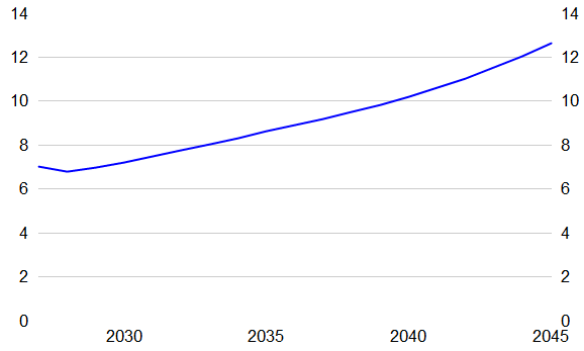


GRC - Greece

Change in net debt from 2027, % pts of GDP

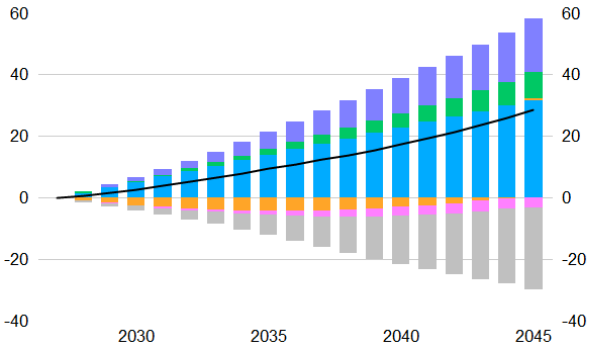


Gross financing needs, % of GDP

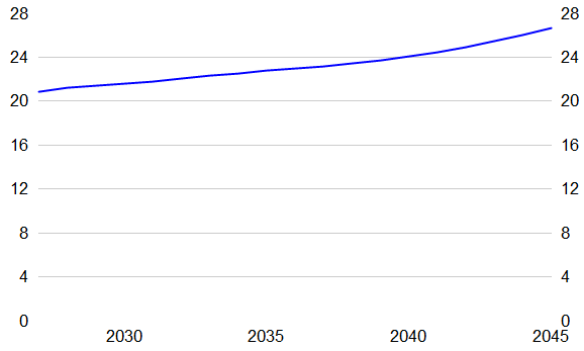


HRV - Croatia

Change in net debt from 2027, % pts of GDP

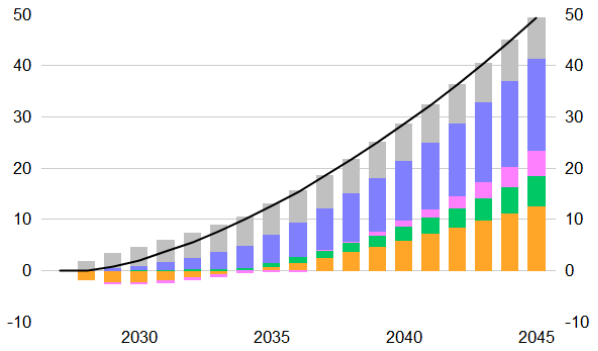


Gross financing needs, % of GDP

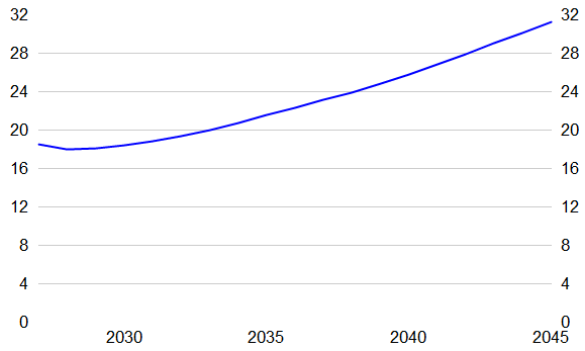


HUN - Hungary

Change in net debt from 2027, % pts of GDP

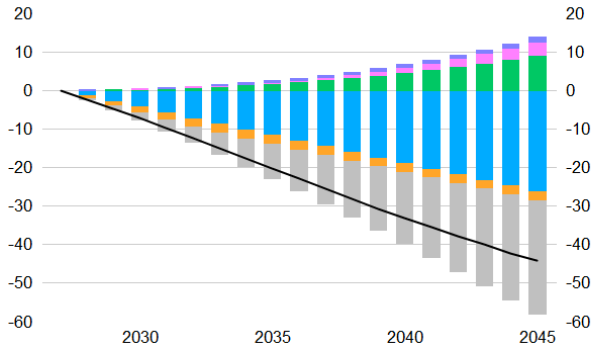


Gross financing needs, % of GDP

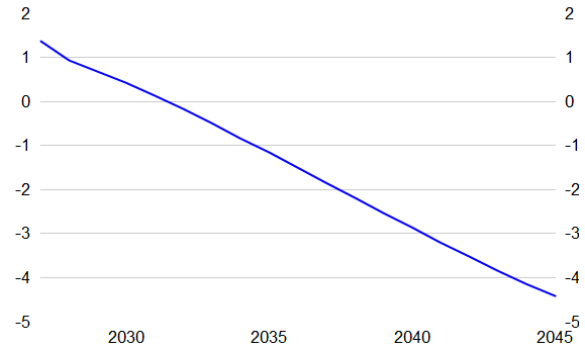


IRL - Ireland

Change in net debt from 2027, % pts of GDP

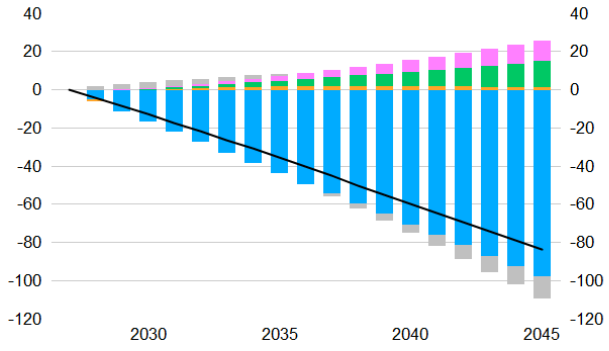


Gross financing needs, % of GDP

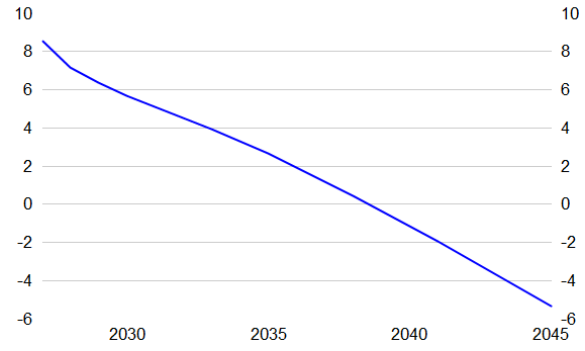


ISL - Iceland

Change in net debt from 2027, % pts of GDP

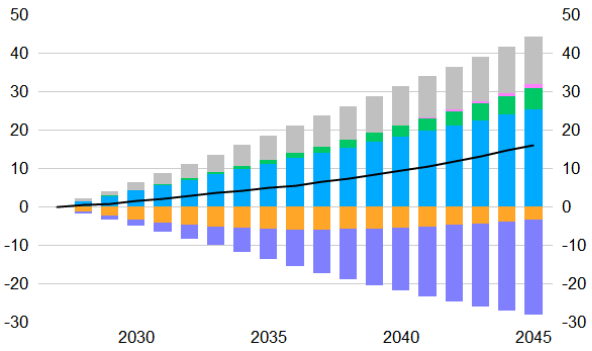


Gross financing needs, % of GDP

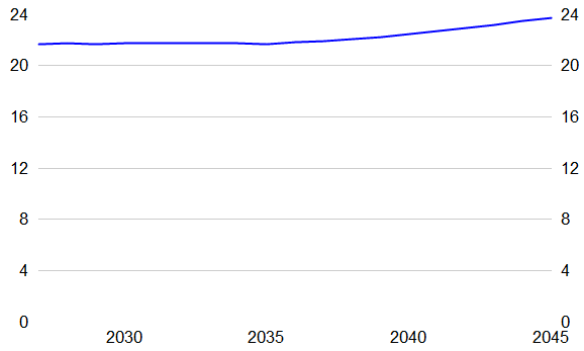


ISR - Israel

Change in net debt from 2027, % pts of GDP

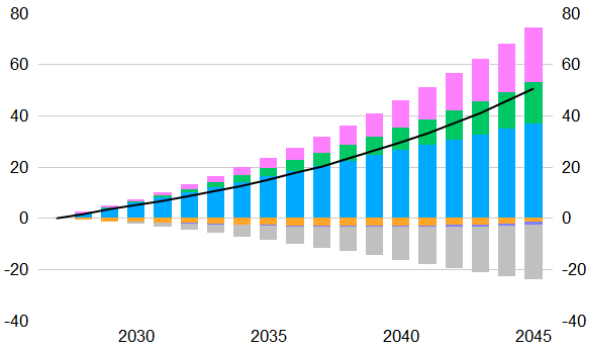


Gross financing needs, % of GDP

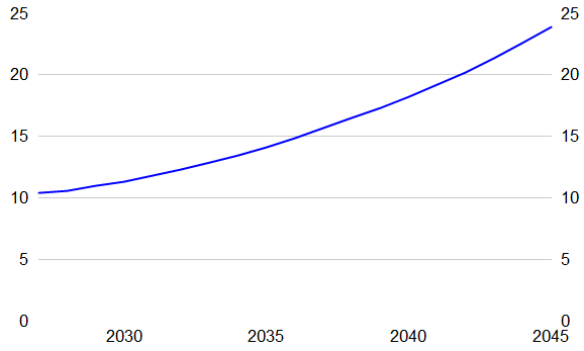


KOR - Korea

Change in net debt from 2027, % pts of GDP

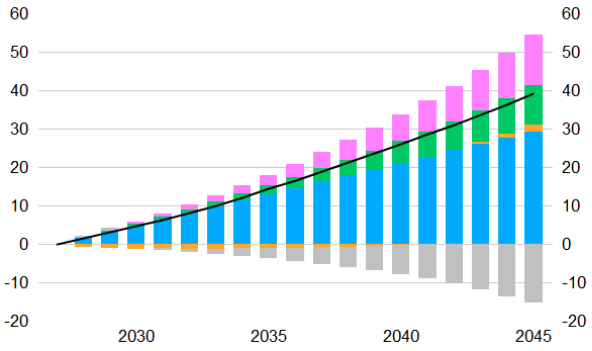


Gross financing needs, % of GDP

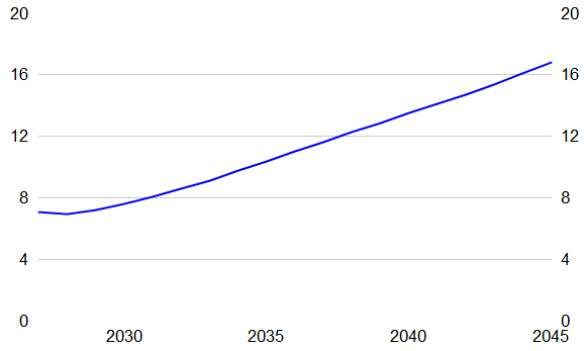


LTU - Lithuania

Change in net debt from 2027, % pts of GDP

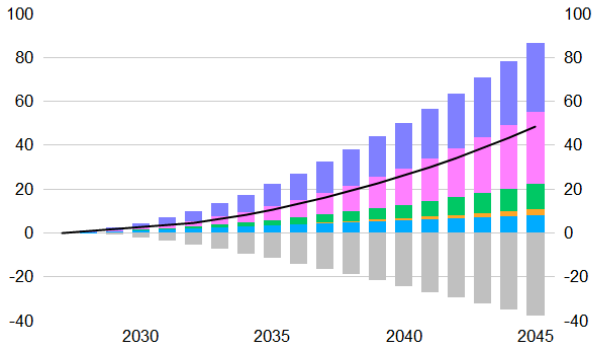


Gross financing needs, % of GDP

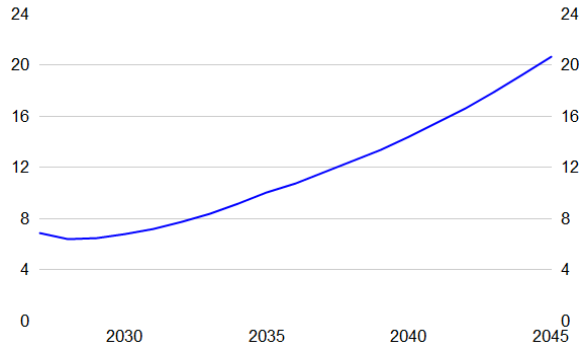


LUX - Luxembourg

Change in net debt from 2027, % pts of GDP

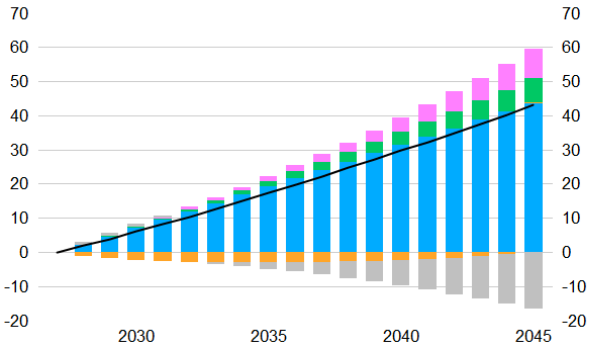


Gross financing needs, % of GDP

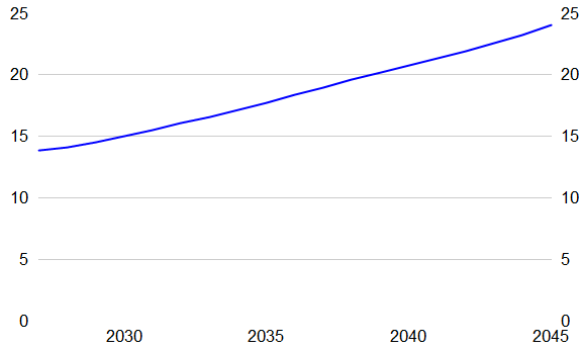


LVA - Latvia

Change in net debt from 2027, % pts of GDP

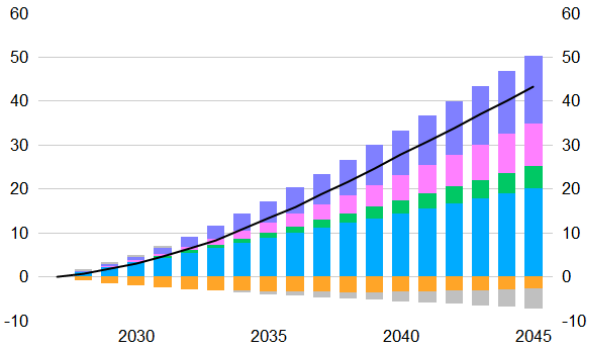


Gross financing needs, % of GDP

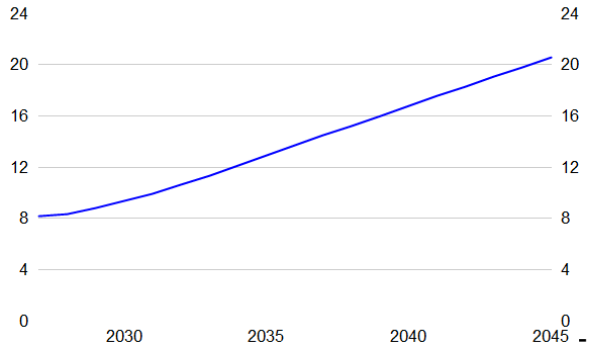


NLD - Netherlands

Change in net debt from 2027, % pts of GDP

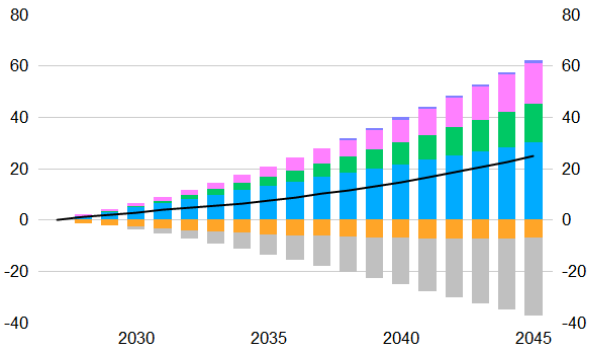


Gross financing needs, % of GDP

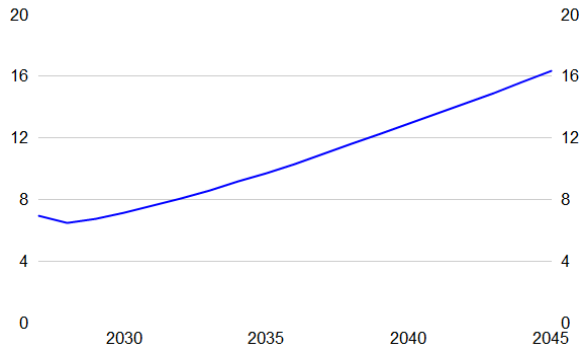


NZL - New Zealand

Change in net debt from 2027, % pts of GDP

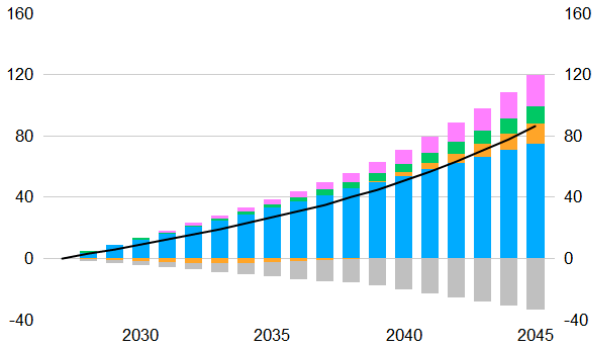


Gross financing needs, % of GDP

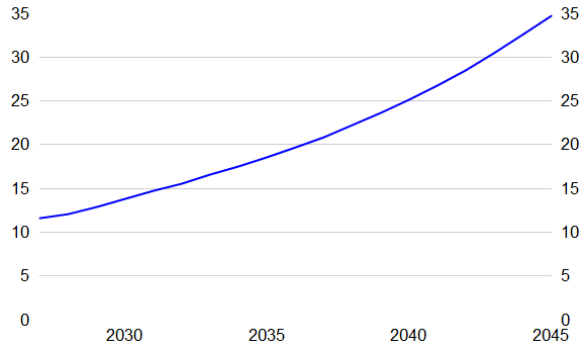


POL - Poland

Change in net debt from 2027, % pts of GDP

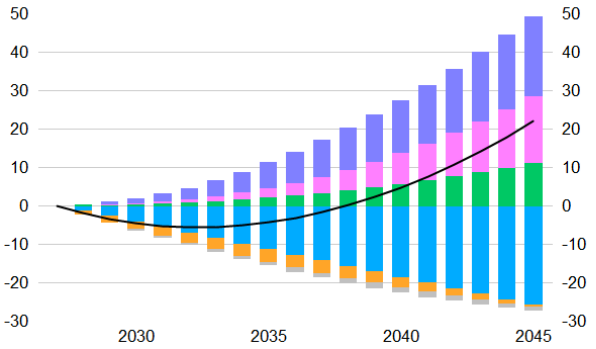


Gross financing needs, % of GDP

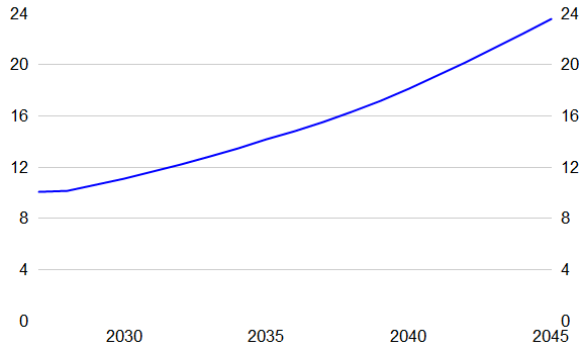


PRT - Portugal

Change in net debt from 2027, % pts of GDP

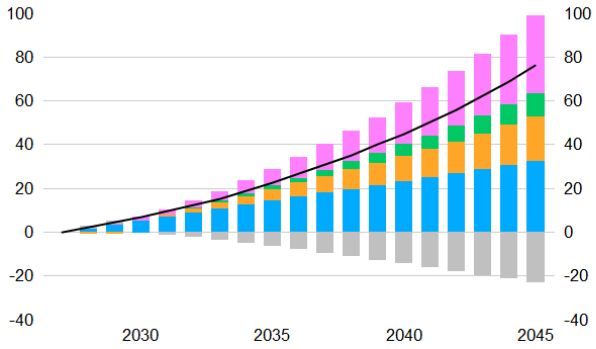


Gross financing needs, % of GDP

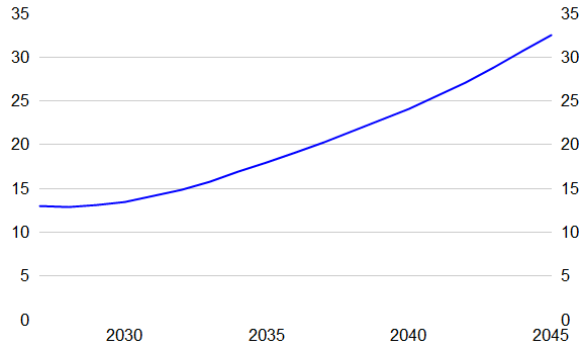


ROU - Romania

Change in net debt from 2027, % pts of GDP

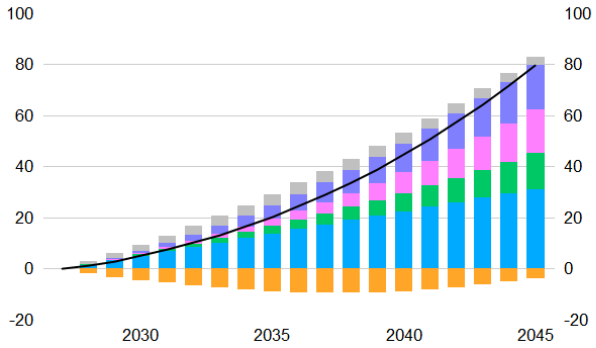


Gross financing needs, % of GDP

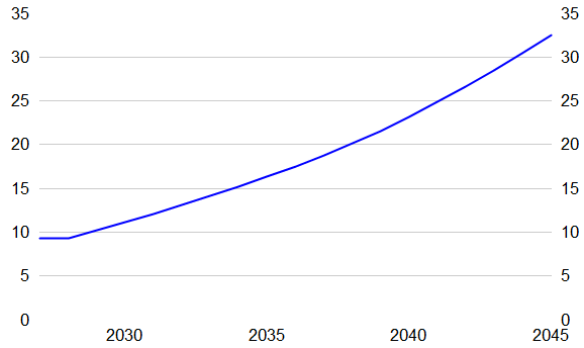


SVK - Slovak Republic

Change in net debt from 2027, % pts of GDP

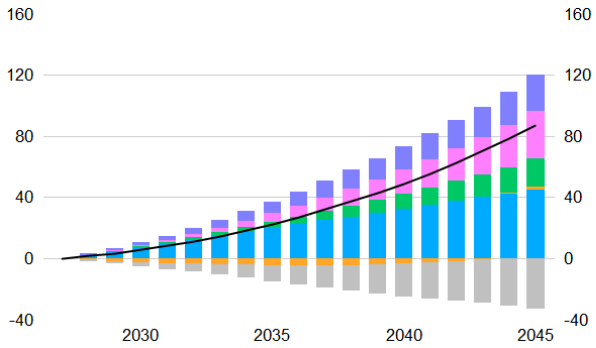


Gross financing needs, % of GDP

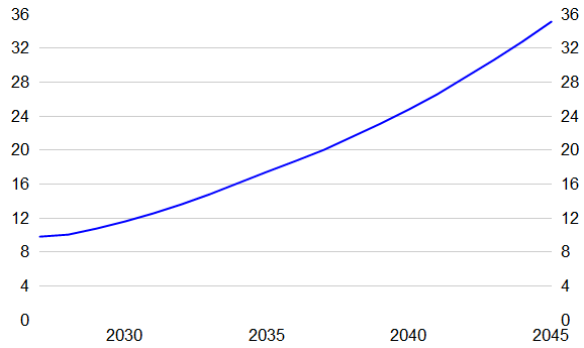


SVN - Slovenia

Change in net debt from 2027, % pts of GDP

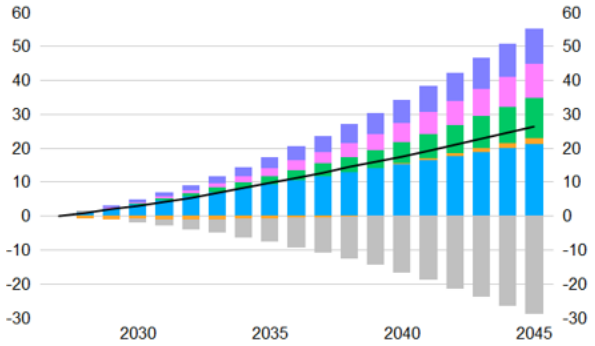


Gross financing needs, % of GDP



SWE - Sweden

Change in net debt from 2027, % pts of GDP



Gross financing needs, % of GDP